Environment Case Study Gate

Figure 5, Drug Exposure: the Hydrolysate Inhibitor Panels, Held for a Decision

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October 19, 2026

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Figure and table status: ● on file, every value in a committed result file ◗ partial, a run unfinished or unscored ❍ idea only, a sketch or nothing run

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1 The decision ✗

Manuscript section R5 covers drug exposure: a compound and its dose as a perturbation the model reads alongside the genotype. The case study is a hydrolysate. Sorghum hydrolysate carries six inhibitors that a fermentation host meets together, and the private Bioscreen runs of 2021 measured all 6 of them on one strain, bAID in aerobic YPD: 180 titration wells, 197 wells over 63 combinations, and 600 wells over 3 isobole grids, 977 wells in all. The models that would predict them were trained on the pooled chemogenomic table: 4 published screens, 6,042,771 cells, 5,463 compounds over 5,863 genes. Nothing here is approved. The sheets are mockups of what the result files support, and the symbol after each panel letter says only whether the values behind it are on file. Color is the source of a predicted value: amber, a measured value, whether a published screen or a Bioscreen well; brick, Bliss independence; lilac, Loewe additivity; wheat, the highest single agent; bronze, the mean of the single-compound profiles; steel blue, a model learned on chemistry features, whose kernel and encoder are written in the label; denim, the cell graph transformer; gray, a gene-mean or permutation control and chance. Hatching is the software growth call and appears only where a panel draws both calls; a plain bar is the served raw-curve call, which is primary. A lighter fill of the same color is a second reading of the same arm, and the panel's legend names it. A label with no bar is an arm that has not run.

What the figure would claim. A model trained on published chemogenomic screens can predict how an industrial host behaves under a mixture of inhibitors it has never seen, and it can do so well enough to rank the combinations a process should avoid. The figure is built to make that claim testable rather than to assert it: the bottom row puts the model beside the two model-free references on the same wells, so a reader sees what the model adds over arithmetic on the single-agent curves, including when the answer is nothing.

The four claims, and where each one stands. First, that growth under a combination departs from independence. On file: over the 63 combinations the Loewe rule reaches a growth AUROC of 0.958 and an accuracy of 0.825, Bliss from the observed singles 0.872 and 0.556, and the highest single agent 0.663; over the 21 combinations that grew, Bliss ranks fitness at Spearman 0.573 with a mean signed error of -0.142 and Loewe at 0.713 with 0.128. Of 15 pairs, 12 are synergistic relative to Bliss, 2 additive and 1 antagonistic. Second, that chemogenomic profile similarity anticipates which pairs deviate. This FAILS as pre-registered: with measured profiles in the matrix the association is 0.204 at a permutation p of 0.462 over 15 pairs, and the all-predicted matrix carries the opposite sign, -0.686 at p 0.006 on raw profiles, which the fingerprint explains rather than the biology. Third, that deletion profiles compose under a mixture by a rule measurable on published data. On file: over 26 Hillenmeyer HET pairs the mean of the singles is the best fixed rule in 24 of them, median R-squared 0.365 against 0.091 for the sum and 0.050 for the highest single, and the free linear fit puts weight 0.529 on one single and 0.593 on the other. Fourth, post-analysis only: composed profiles

nominate the genes a screen under each combination would recover, 68 core genes from the best-available matrix and 105 from the all-predicted one. No screen exists to test them.

The bars a model has to clear. Growth: AUROC 0.958, the Loewe rule from the ex21 Hill fits. Fitness over the combinations that grew: Spearman 0.573, Bliss from the observed ex23 singles. Both are model-free, both come from the same wells the model would be scored on, and both are already in a committed result file. On the published side the bar is the nested ridge on fingerprint counts at a median centered Spearman of 0.359 over 96 compound-cold folds, which the next best arm of the ladder trails by a hundredth (0.357).

The angle, after ten rounds. The figure is not a better chemogenomic model. Nested ridge on fingerprint counts holds the published bar (0.359) and no transformer arm of any round reaches it. The angle is that the environment is a perturbation of the same kind as a deletion, and one cell model composes both. Once every record is a cell, a deletion strain in a drug and a wild-type host in a mixture being one record type read through one head, and a mixture is read one inhibitor at a time and composed as Bliss independence, the model trained on the published deletion screens and two anchors orders the 63 private combinations at a growth AUROC of 0.930 without a titration well in training, above the Bliss rule on the measured singles (0.872) and 0.942 with the Hoepfner and Hillenmeyer screens added, at Loewe's level (0.958). The same compositional algebra as the gene arm follows: singles first, with the dose offset giving them a real dose axis (titration Spearman 0.869 at growth AUROC 0.851), then a deviation term, which is the environment's analog of a genetic interaction and the open piece; every dose arm overpredicts the isobole interiors, which is what a missing interaction term looks like. Pooled published data buys transfer, not benchmark points: the screens do not lift the target (0.265 with one source, 0.194 with four, inside one spread) and are not learned at a held-out compound, yet the zero-shot private readout rises with every screen added, three runs per arm and the steps inside the spread, a measured ordering rather than a cleared difference. The private wet-lab data is the environment arm's held-out level, the way the held-out query pairs are the gene arm's: a readout, a culture and a dose regime no training record carries.

Where the full record is. The sheets draw the figure; the measurements behind them are written up elsewhere, and the source table under each figure points to the file. `notes-tex/wet-lab/040-inhibitor-synergy-wetlab` is the case-study document: the 2021 runs, the growth calls, the four claims scored, and the confounds each caption carries. `notes-tex/metabolism/031-betaxanthin-module` and `notes-tex/metabolism/032-amino-acid-module` hold the campaign whose panel conventions this document copies. The chemogenomic side is in three experiment records rather than documents: `experiments/033-env-chemgen-pooled` is the pooled table and its axes, `experiments/035-env-chemgen-vanacloig-cgt` the first compound-cold campaign and the learning curve, and `experiments/038-env-chemgen-vanacloig-cgt-corrected` the ladder and the factorized rounds on the corrected store. The 035 and 038 branches are not landed, so the panel script reads their result files by absolute path and records every path it read in `experiments/040-inhibitor-synergy-wetlab/results/figure_gate_manifest.json`.

What a decision here settles. Whether the figure is the fifteen panels of fig. 2 or the three-by-three the experiment note proposes, which of the fifteen panels earns main-text space against supplementary space, and whether the bottom row ships with the model beside the rules or with the rules alone. Nothing in this document is a publication-ready panel; the panels are drawn at Nature print width so that the argument, not the artwork, is what gets judged.

2 The mockups ✗

Each caption gives, per panel, what is plotted, the statistic and the status; the table under each figure gives the files behind it. The statistic on the published side is always the per-compound centered Spearman under compound-cold folds, median over compounds, with 26 compounds fitted per fold; on the wet-lab side it is the growth AUROC, the area under the ROC curve of a predicted fitness as the score for whether a combination grew or not under the served call, over the 63 combinations (a rule or a model is read on its fitness, never on a probability, so the AUROC is the one scale-free number the two share), and the fitness Spearman over the 21 that grew. Color is the source of a predicted value: amber, a measured value, whether a published screen or a Bioscreen well; brick, Bliss independence; lilac, Loewe additivity; wheat, the highest single agent; bronze, the mean of the single-compound profiles; steel blue, a model learned on chemistry features, whose kernel and encoder are written in the label; denim, the cell graph transformer; gray, a gene-mean or permutation control and chance. Hatching is the software growth call and appears only where a panel draws both calls; a plain bar is the served raw-curve call, which is primary. A lighter fill of the same color is a second reading of the same arm, and the panel's legend names it. A label with no bar is an arm that has not run.

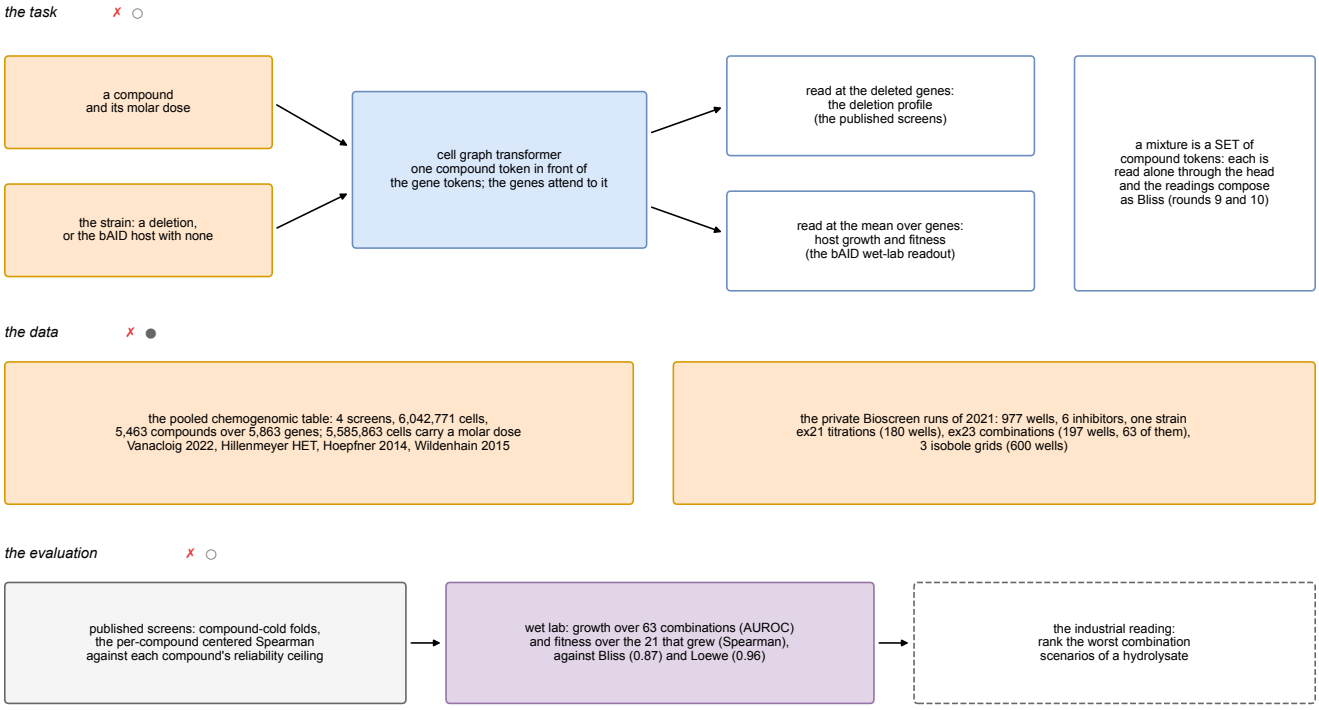


Figure 1: The task, the data and the evaluation. Three blocks, each a layout rather than a drawn result; a dashed box is proposed and not run. **The task ✗○**: a compound and its molar dose enter the cell graph transformer as one token placed in front of the gene tokens, which attend to it; a published-screen strain is read at its deleted genes, giving a deletion profile, and the bAID host, which carries no deletion, is read at the mean over genes, giving growth and fitness. A mixture is a set of compound tokens, one per inhibitor: each is read alone through the shared head and the readings compose as Bliss independence (rounds nine and ten), which is what took the private readout from chance to the rule's level. **The data ✗●**: the pooled chemogenomic table over 4 screens, 6,042,771 cells and 5,463 compounds across 5,863 genes, of which 5,585,863 cells carry a molar dose, beside the private Bioscreen runs, 977 wells over 6 inhibitors on one strain. **The evaluation ✗○**: compound-cold folds and the per-compound centered Spearman on the published side, growth and fitness against Bliss and Loewe on the wet-lab side, and the industrial reading, which is to rank the worst combination scenarios of a hydrolysate.

Sources for fig. 1	
the task	Nothing drawn from a file. The encoder it describes is the environment-encoder head of `038/`; the host head is the proposed adaptation of `experiments/040-inhibitor-synergy-wetlab/scripts/train_mixture.py`. Status: idea.
the data	`033/pool_totals.csv`, `033/store_axes.csv` (the screens, the cells, the compounds, the genes, the molar-dose coverage); `wetlab_wells.csv` and `wetlab_provenance.json` (the wells, the runs, the raw-mirror shas). Status: on file.
the evaluation	`038/ladder/ladder_r2_scores.csv` (the fold size); `mixture_scores.csv` (the two bars). The boxes themselves are a layout, so the block is marked idea even though its counts are on file.

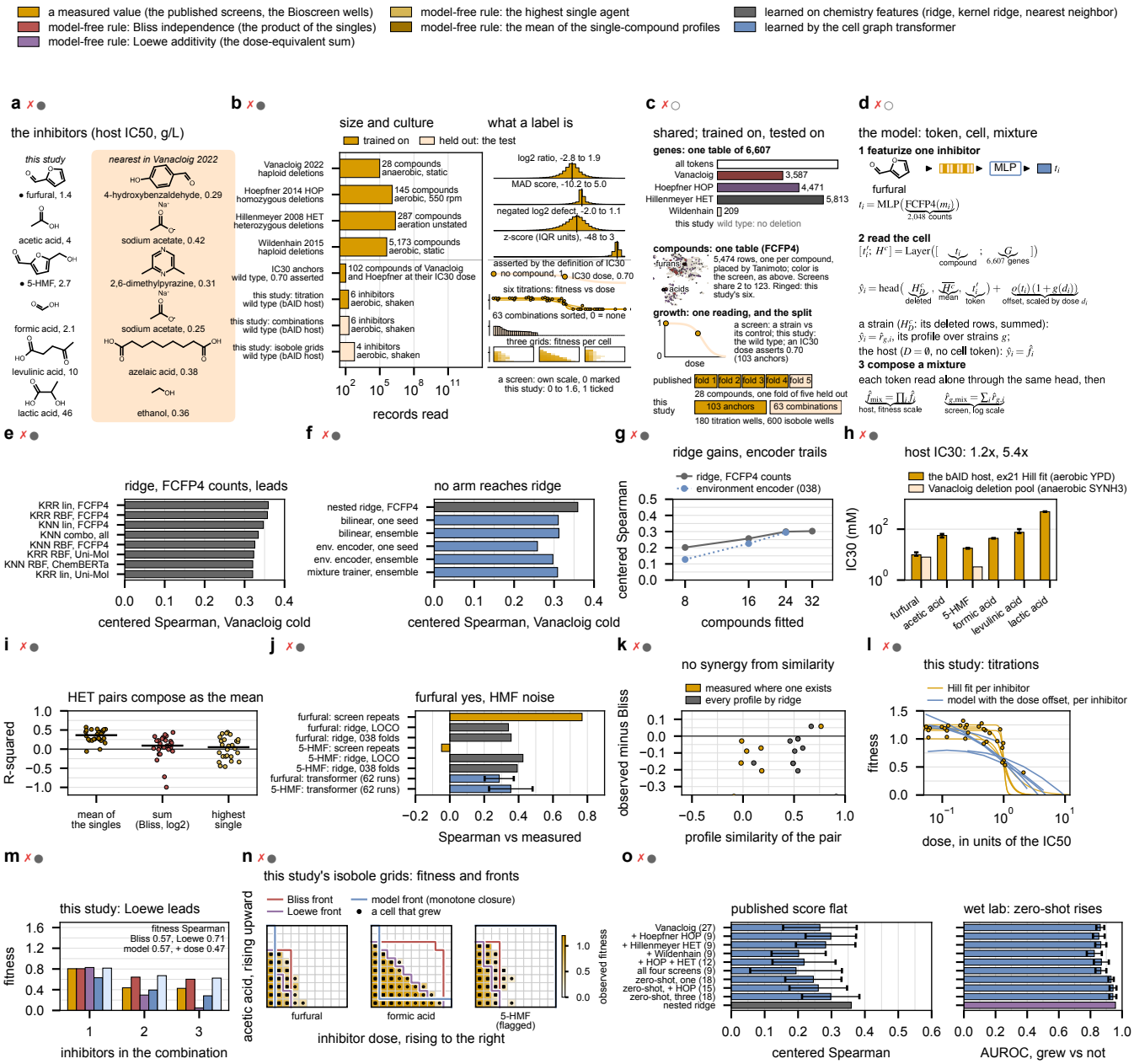


Figure 2: Composition is what the wet lab buys, and no published screen lifts the published score. The six hydrolysate inhibitors are met by one wild-type host; Vanacloig 2022 profiles two of them itself, and the nearest profiled compound of each of the other four sits beside it (**a**). Four published deletion screens of three genotypes and this study's wild-type wells differ in size, culture and in what a label is, a strain-level score on a screen's own scale against a growth rate per well (**b**); what joins them is one gene table, one compound table and one growth reading, with a screen's IC30 dose asserting the wild type at 0.700 (**c**). The cell graph transformer reads one token per inhibitor through one head, the host as the empty genotype at the gene mean, the dose as a gain on the medium's offset, and composes the readings as Bliss independence (**d**). On the published side nested ridge on fingerprint counts, 0.359, is not beaten by any transformer arm (**e**, **f**), ridge gains about 0.049 per doubling of fitted compounds with the environment encoder below it at every count (**g**), and adding screens leaves the compound-cold score flat (**o**, left). On the wet-lab side Loewe additivity from the titrations orders the 63 combinations at a growth AUROC of 0.958 and Bliss from the measured singles at 0.872; the model trained on the anchors and the titration reads 0.862 on combinations it never saw and its front sits inside the isobole grids (**m**, **n**), and the zero-shot arms, which never see a titration well, rise with every screen added, inside the run spread (**o**, right). Three readings qualify the join. The host needs 1.23 and 5.42 times the screen's IC30 of furfural and 5-HMF (**h**); a screen's label is a strain's growth against its own control on its own scale, this study's a wild-type growth rate with no growth in a share of 0.650 of the combination wells (**b**); and the published profiles do not reach the private question: pairs compose as the mean of their singles (**i**), furfural's profile is predicted at 0.341 where its screen repeats at 0.769 while 5-HMF's screen does not repeat (**j**), and the similarity of two inhibitors' profiles does not anticipate their pair's departure from Bliss, Spearman 0.204 at p 0.462 (**k**). The titrations themselves (**l**) are read by the model only with the dose offset, Spearman 0.890 against 0.452 without it. Color is the source of a value (key above the sheet); dark and light fills in **b** and **c** are trained on and held out. Public scores are compound-cold centered Spearman, median over held-out compounds; wet-lab scores are the served growth call.

Panels of fig. 2. **a**, The six inhibitors from their curated SMILES at the ACS 1996 drawing standard at half the standard's bond length (0.19 cm; a 0.6 pt line, double bonds spaced at 18 percent of the bond, Arial labels, implicit hydrogens on the heteroatoms only; RDKit's ACS mode through `torchcell.utils.acs_structure`), this study's six in one column and, on the measured-value field beside each, the Vanacloig 2022 compound nearest it by FCFP4 count Tanimoto over the pool's curated SMILES, the inhibitors themselves excluded, with the similarity after the name; a dot marks furfural and 5-HMF, which the screen profiles itself. 4-hydroxybenzaldehyde to furfural (0.286), 2,6-dimethylpyrazine to 5-HMF (0.306), sodium acetate to acetic acid (0.417) and to formic acid (0.250), azelaic acid to levulinic acid (0.375), ethanol to lactic acid (0.357). The hydrolysate's two families are the furan aldehydes from sugar dehydration (pentoses to furfural, hexoses to 5-HMF) and the weak acids (5-HMF opens to levulinic and formic acid, acetyl groups release acetic acid). After each inhibitor's name, the host's half-maximal dose from its titration fit. ✗● **b**, Left, what the mixture trainer reads from a four-source run, one bar per source on a log scale: a screen's cells, this study's wells, the anchors' conditions, with the genotype on each row's second line (Vanacloig's haploid deletions on the PDR1, PDR3 and SNQ2 sensitized host; Hoepfner's homozygous and Hillenmeyer's heterozygous deletions in diploids; Wildenhain's haploid deletions; this study's wild-type BY4742-iAID6 host with no deletion) and the compound count and the culture beside each bar. The task's profiles are 97,772 anaerobic, static cells over 28 compounds; the auxiliary screens add 1,234,523, 2,345,856 and 406,460 aerobic cells, 4,084,611 in all. The host readout is aerobic, shaken YPD, a different oxygen regime from the profiles. The anchors are the 103 published conditions whose served dose is an IC30 (102 compounds of Vanacloig and Hoepfner, and the compound-free medium), read as host records at a fitness of 0.700 by the definition of IC30, the dose that inhibits the wild type's growth by 30 percent, and at 1 for the compound-free medium: asserted from the screens' dosing, never measured in this study (`scripts/mixture_data.py`, `anchor_records`). Right, what a label is. A screen's row is a histogram of the value the trainer reads before it standardizes the source: one deletion strain's growth under one compound at one dose, relative to that screen's control, oriented so that a sick deletion is negative, on the screen's own scale (Vanacloig a log2 ratio to the inhibitor-free control, Hoepfner a MAD-scaled sensitivity score, Hillenmeyer a negated fitness-defect log2 ratio, Wildenhain a z-score in IQR units), the band spanning its 0.5th to 99.5th percentile with zero marked. This study's rows draw the labels as they are, on one fitness scale of 0 to 1.6 with 1 ticked: the six titration curves with their wells, the 63 combinations as bars sorted by fitness with no growth at zero, and the three isobole grids as cells; no growth in a share of 0.350 of the titration wells, 0.650 of the combination wells and 0.710 of the isobole wells. The anchors' row is a sketch of what is asserted and never measured: fitness 1 with no compound and 0.700 at a screen's IC30 dose. The anchors and the titration are trained on; the combinations and the isobole grids never. ✗● **c**, What the sources share and what is trained on and tested on, as three drawings. Genes: every screen is a deletion collection of the same genes, so one table of 6,607 gene tokens serves all of them, each screen's coverage drawn as a bar of that length (209 to 5,813); this study's host deletes none. Compounds: every compound is one row of one structure table (FCFP4 counts), and the scatter is that table, all 5,474 rows placed in two dimensions by UMAP on the Tanimoto similarity between their fingerprints taken as bit sets. A point's color is the screen that holds it, read off the gene bars above: Wildenhain, which holds most of the table, is the pale ground, the three smaller screens are drawn dark over it, and gray is a row no screen dosed. The screens spread over the whole table and share 2 to 123 compounds with each other, and this study's six, ringed, sit inside it in two labeled places: furfural and 5-HMF among the furans, which Vanacloig also holds, and the four acids, which no screen dosed. Growth: every value is a growth reading, a screen's of a strain against its control and this study's of the wild type, and an IC30 dose asserts the wild type at 0.700, the marked point of the sketch, which is what ties a screen's dose scale to the host's. The split, in the fills of b: the trainer sees every screen, with Vanacloig in compound-cold folds (one of five held out, 28 compounds), the 103 anchors and the titration (55 conditions, 180 wells); it is tested on the held-out compounds' profiles (centered Spearman over genes) and on this study's 63 combinations (growth AUROC, fitness Spearman) and 600 isobole wells (the growth front). A layout; the counts are the trainer's. ✗○ **d**, The model in three stages, its equations the trainer's (`MixtureModel` of `scripts/train_mixture.py`), set by LaTeX with each component named under its brace. One inhibitor m_i (furfural, drawn as in a) is featurized as its 2,048 FCFP4 counts and passed through the compound MLP to one token t_i. The token is placed in front of the 6,607 gene tokens G, a learned table read through the one-layer cell graph transformer and trained with the rest at its own rate, not the composite sequence embedding of the interactions figure; one further transformer layer reads the sequence, so the token reads the genome and the genes read the compound, giving the cell-in-medium state H^c and the read token t_i'. One head reads a state made of the rows of H^c at the strain's deleted genes D, the mean over genes and t_i' (and, not drawn, the encoder's own post-deletion rows and a learned source token), scaled and shifted per source; a strain gives $\hat{r}_{g,i}$, its response, which over strains g is the profile the centered Spearman ranks; the host, which deletes nothing, gives $\hat{f}_i$, its growth. There is no cell token: the host's state is the gene mean. The dose enters once, as a gain $1 + g(d_i)$ on the medium offset $o(t_i)$, a per-medium bias added to the head, with g an MLP of the standardized log molar dose started at zero (the dose offset of round ten). A mixture is a set of tokens, each read alone through that head, and the readings compose as Bliss independence: a product on the host's fitness scale, a sum on a screen's log response. The model is then scored on the 63 combinations it never saw. A layout; the counts are the trainer's. ✗○ **e**, The ladder, ten best arms of 67 on Vanacloig's compound-cold folds: nested ridge on fingerprint counts reads 0.359, the next arm 0.357, the tenth 0.320. ✗● **f**, Which model predicts an unseen Vanacloig compound's profile. The second round is complete: over 96 compound-evaluations, 32 compounds on each of three fold seeds, the three-seed ensemble of the bilinear control

reads 0.312 and of the environment encoder 0.297, both below nested ridge at 0.359. Paired on compound and fold, the encoder minus its control is -0.036 with 37 of 96 up. The last bar is the mixture trainer's single-source arm on its own basis (28 compounds with a molar dose, folds 0 to 2, the three-seed ensemble): 0.309. ✗● **g**, The learning curve in fitted compounds. Ridge: 0.202 at eight to 0.303 on the pool, about 0.049 per doubling. The environment encoder (038 round 17, 96 compound-evaluations per point): 0.127, 0.226 and 0.295 at eight, sixteen and twenty-four, below ridge at every count and closing on it. ✗● **h**, Dose, the host's potency against the screen's: furfural 9.82 millimolar against 8, ratio 1.23; 5-HMF 17.9 against 3.30, ratio 5.42; the other four are in no screen. ✗● **i**, How a pair's profile composes from its singles over 26 Hillenmeyer HET pairs: the mean is best in 24, median R-squared 0.365 against 0.091 for the sum and 0.050 for the highest single. ✗● **j**, The predicted profile against the measured one. Furfural: the screen repeats itself at 0.769, ridge reads 0.341; 5-HMF: -0.048, which is noise, against 0.425. The two profiles share 0 significant terms. The cell graph transformer, every round-3 and round-4 run on both machines in which the compound was held out: furfural 0.287 over 62 runs, 5-HMF 0.355 over 62, with the run spread as the whisker. ✗● **k**, The pre-registered test, which FAILS: does the similarity of two inhibitors' profiles anticipate how far their pair departs from Bliss. With measured profiles where one exists the association is 0.204 at a permutation p of 0.462 over 15 pairs; with every profile predicted by ridge it is -0.225 at p 0.425. ✗● **l**, This study's six titrations with their Hill fits, dose in units of each compound's own half-maximal dose; slopes run 1.21 to 20.0. The model whose host, the empty genotype read through the shared cell head, is trained on the published anchors alone reads a flat line at the anchors' assumed fitness for every inhibitor at every dose, Spearman -0.222 against the 54 observed single-agent conditions (not drawn); trained on these wells it fits, Spearman 0.452, and is still nearly flat in dose, because without a dose channel the host reads one value per compound (not drawn). Drawn, the same model with the dose offset (round ten): the compound tokens stay dose-free and the dose scales the medium offset, so the curve falls with dose, Spearman 0.890 on these wells. ✗● **m**, This study's combinations that grew, observed against the two rules, by number of inhibitors. Growth AUROC: Loewe 0.958, Bliss 0.872. Fitness Spearman: Loewe 0.713, Bliss 0.573. The fourth bar is the model (single source, cell readout, trained on the anchors and the titration, mean over 20 runs on 5 folds, seeds and machines), which reads a medium one inhibitor at a time through the shared head and composes the readings as Bliss independence: its predicted fitness falls with the inhibitor count, growth AUROC 0.862, fitness Spearman 0.566. The fifth bar, lighter, is the same model with the dose offset (3 runs): it reads the titration (l) and still composes, growth AUROC 0.849, fitness Spearman 0.466, with the fitness at one inhibitor at the observed level and a shallower fall at two and three. Under mean pooling of the inhibitor tokens the same model read one fitness at every count (AUROC 0.535). ✗● **n**, This study's 3 isobole grids as observed fitness, white at none, each grid's inhibitor named on its lower edge and acetic acid on every grid's left edge, with the cells that grew marked and each rule's front drawn where its predicted fitness crosses the growth threshold: Bliss in red, Loewe in purple, every front a solid line over a white halo. Mean observed minus Bliss over the 81 interior cells: -0.082, -0.446 and -0.017, the last a single run with growth islands and flagged. In blue, the Bliss-composed model's front after the monotone closure, inside the grid for the first time: the model's mean observed minus predicted over the interior cells is -0.084, +0.016 and -0.102, against the rule's -0.082, -0.446 and -0.017, so on the formic acid grid, where the rule on measured singles overpredicts by 0.45, the model is within 0.02. The dose-offset arm is scored on the same grids (supplement eight) and not drawn. A caveat the sheet carries: the model's single-agent curves are nearly flat in dose (l), so a product of two low readings matching a low interior is not yet evidence that it reads the interaction; the closure drops 0 cells on the furfural grid. ✗● **o**, The published screens added to the task under the Bliss cell readout, every fitted and zero-shot arm on the same rows, the mean over every run on fold seed 0 with the run spread as the whisker. Left, the compound-cold centered Spearman: one source 0.265 (27 runs), with Hoepfner HOP 0.298, with Hillenmeyer HET 0.283, with Wildenhain 0.202, three 0.218, four 0.194, every one inside one spread and below nested ridge at 0.359; no screen lifts the task. Right, the growth AUROC of the same arms, each arm's predicted fitness scored against whether each of the 63 combinations grew. The fitted arms, which train on the titration, read 0.862 with one source to 0.866 with four. The zero-shot arms, which never see a titration well, read 0.930, 0.943 and 0.942 on one, two and three sources (18 runs each), against Loewe at 0.958: a rise with every screen added, with the steps inside the spread, so a measured ordering rather than a cleared difference. ✗●

Sources for fig. 2

a `profile_fingerprint_check.csv` (the curated SMILES of the pool's compounds; the sheet computes each inhibitor's nearest by FCFP4 count Tanimoto, itself and the other inhibitors excluded) and `source_inhibitor_overlap.csv` (which inhibitors a screen profiles), and `scripts/mixture_data.py` `IDENTITY_GAP_SMILES` for formic and lactic acid; `single_agent_fits.csv` for the half-maximal doses. The structures are RDKit's ACS 1996 drawings as vector paths (`torchcell.utils.acs_structure`). Status: on file.

b `results/mixture-delta/r3_vanhophetwil_e150_p0/*_f0_counts.csv`, the per-source counts `scripts/train_mixture.py` writes beside every run; `source_label_histograms.csv` and `source_label_summary.csv` (`scripts/source_labels.py`, the oriented value per source read from the 033 cell table and from `wetlab_wells.csv`); `wetlab_wells.csv`, `single_agent_fits.csv`, `mixture_combinations.csv` and `isobole_cells.csv` for this study's rows drawn as they are. The genotypes and the culture conditions are the loaders' sourced values (`vanacloig2022.py`, `hoepfner2014.py`, `hillenmeyer2008.py`, `wildenhain2015.py`, `private_torchcell/bioscreen.py`; the Bioscreen archive states no shaking, so "shaken" is the operator's statement). "This study" is the private Bioscreen archive: the titration is run ex21, the combinations ex23, the isobole grids ex26 to ex28. Status: on file.

c the same counts file; `source_agreement.csv` (`scripts/source_axes.py`, the compounds two screens share, from 033's cross-source cells); `source_inhibitor_overlap.csv` (`scripts/source_labels.py`, the inhibitors a screen holds, matched on InChIKey); `scripts/mixture_data.py` `anchor_records` for the anchors. The drawing is a layout. Status: idea.

d `scripts/train_mixture.py` (the architecture, `MixtureModel`) and `scripts/mixture_data.py` (the fingerprint); the drawing is a layout. Status: idea.

e `038/ladder/ladder_r2_summary.csv`, centered target, full compound count. Status: on file.

f `038/ladder/ladder_r2_summary.csv`; `038/factorized/r9_control/*_scores.csv`; `038/factorized/r10_envenc/*_scores.csv`; `results/mixture/r3_van_e150/*_scores.csv` for the mixture-trainer bar. Status: on file.

g `035/learning_curve/ridge_curve.csv`, centered target; `038/factorized/r17_curve/*_scores.csv` for the encoder. Status: on file.

h `single_agent_fits.csv` (the host, with bootstrap intervals from `single_agent_bootstrap.csv`); `vanacloig_ic30.csv` (the screen). Status: on file.

i `het_rule_fit.csv`, `het_summary.json`. Status: on file.

j `profile_predicted_vs_measured.csv` (`loco_32`, centered); `go_overlap_measured_vs_predicted.csv`; every `results/mixture*/r[34]*/*_vanacloig_compounds.csv` on both machines for the transformer. Status: on file.

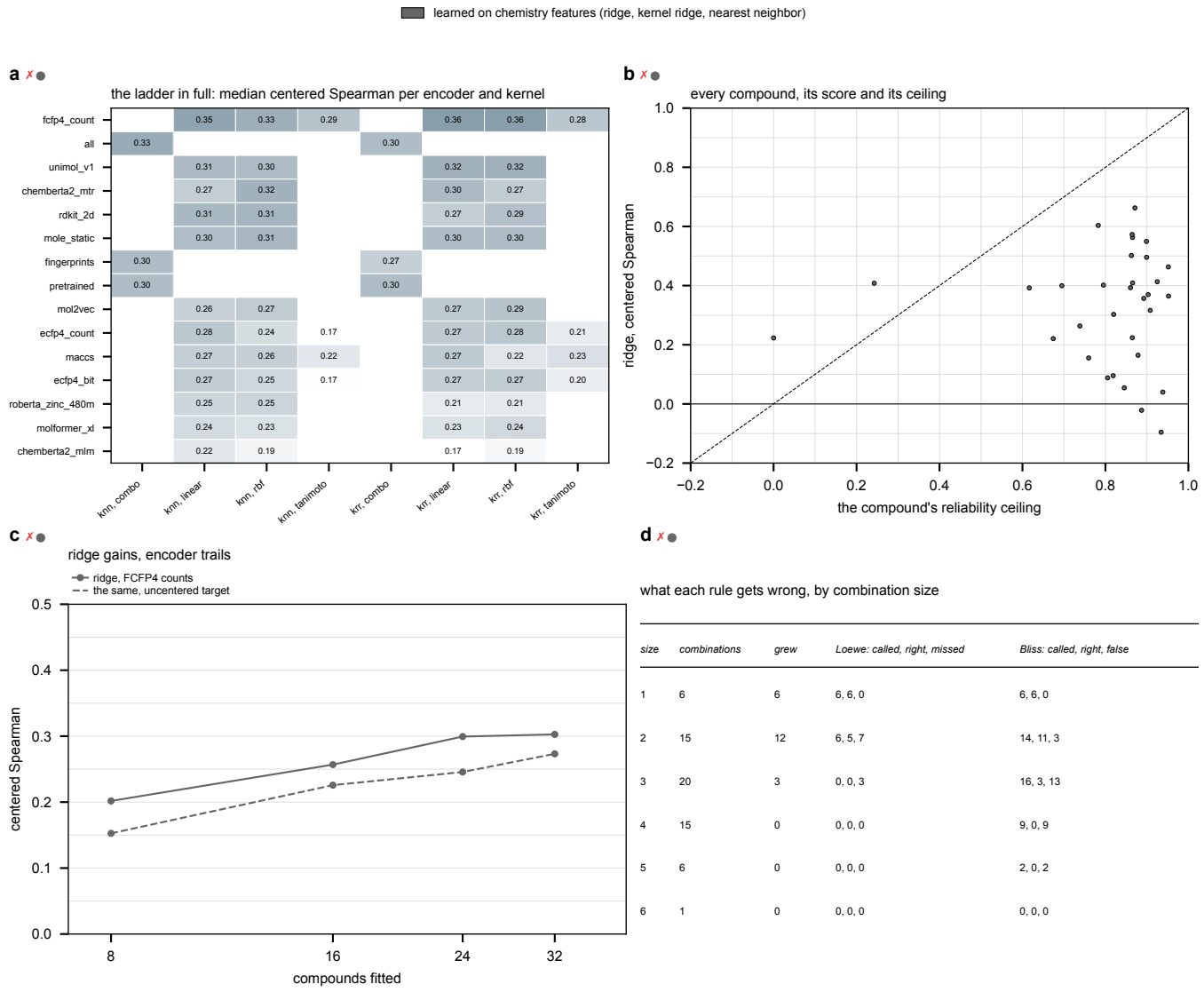
k `similarity_vs_deviation_pairs.csv`, `similarity_vs_deviation.csv` (both profile versions, the served growth call). Status: on file.

l `wetlab_wells.csv` (run ex21), `single_agent_fits.csv`, `single_agent_bootstrap.csv`; `results/host_predictions.csv` (`scripts/r3_host_predictions.py`) for the model curves. Status: on file.

m `mixture_combinations.csv`, `mixture_scores.csv` (run ex23, served call); `results/host_predictions.csv` for the model bars. Status: on file.

n `isobole_cells.csv`, `isobole_summary.csv` (runs ex26 to ex28); `results/host_predictions.csv` and `mixture_rules.monotone_alive` for the model front. Status: on file.

o every `results/mixture*/r10_*_cell_bliss*/*_scores.csv` on both machines (fold seed 0): the `centered_spearman_median` rows on the held-out compounds (left) and the `growth_auroc` rows on the served call of the ex23 combinations (right), with the anchors-only arms beside the fitted ones. Status: on file.



size	combinations	grew	Loewe: called, right, missed	Bliss: called, right, false
1	6	6	6, 6, 0	6, 6, 0
2	15	12	6, 5, 7	14, 11, 3
3	20	3	0, 0, 3	16, 3, 13
4	15	0	0, 0, 0	9, 0, 9
5	6	0	0, 0, 0	2, 0, 2
6	1	0	0, 0, 0	0, 0, 0

Figure 3: Supplement one: the published benchmark in full. a, Every ladder arm, median centered Spearman per encoder and kernel, 62 fitted cells over 15 encoders. Fingerprint counts lead every kernel they are defined for. ✗● **b**, Each of the 32 compounds, its ridge score against its own reliability ceiling: median 0.367, 2 above the ceiling and 2 below zero. ✗● **c**, The learning curve on both targets, the uncentered one dashed. The centered curve is the one the main sheet carries. ✗● **d**, What each rule gets wrong, by combination size: how many combinations the rule called alive, how many of those were right, and what it missed or invented. Loewe misses all 3 growing triples and 7 of 12 growing pairs; Bliss calls 13 of 17 dead triples alive. ✗●

Sources for fig. 3

a	`038/ladder/ladder_r2_summary.csv`. Status: on file.
b	`038/ladder/ladder_r2_scores.csv`, kernel ridge on fingerprint counts, centered, averaged over fold seeds. Status: on file.
c	`035/learning_curve/ridge_curve.csv`. Status: on file.
d	`mixture_scores.csv`, served call, one row per combination size. Status: on file.

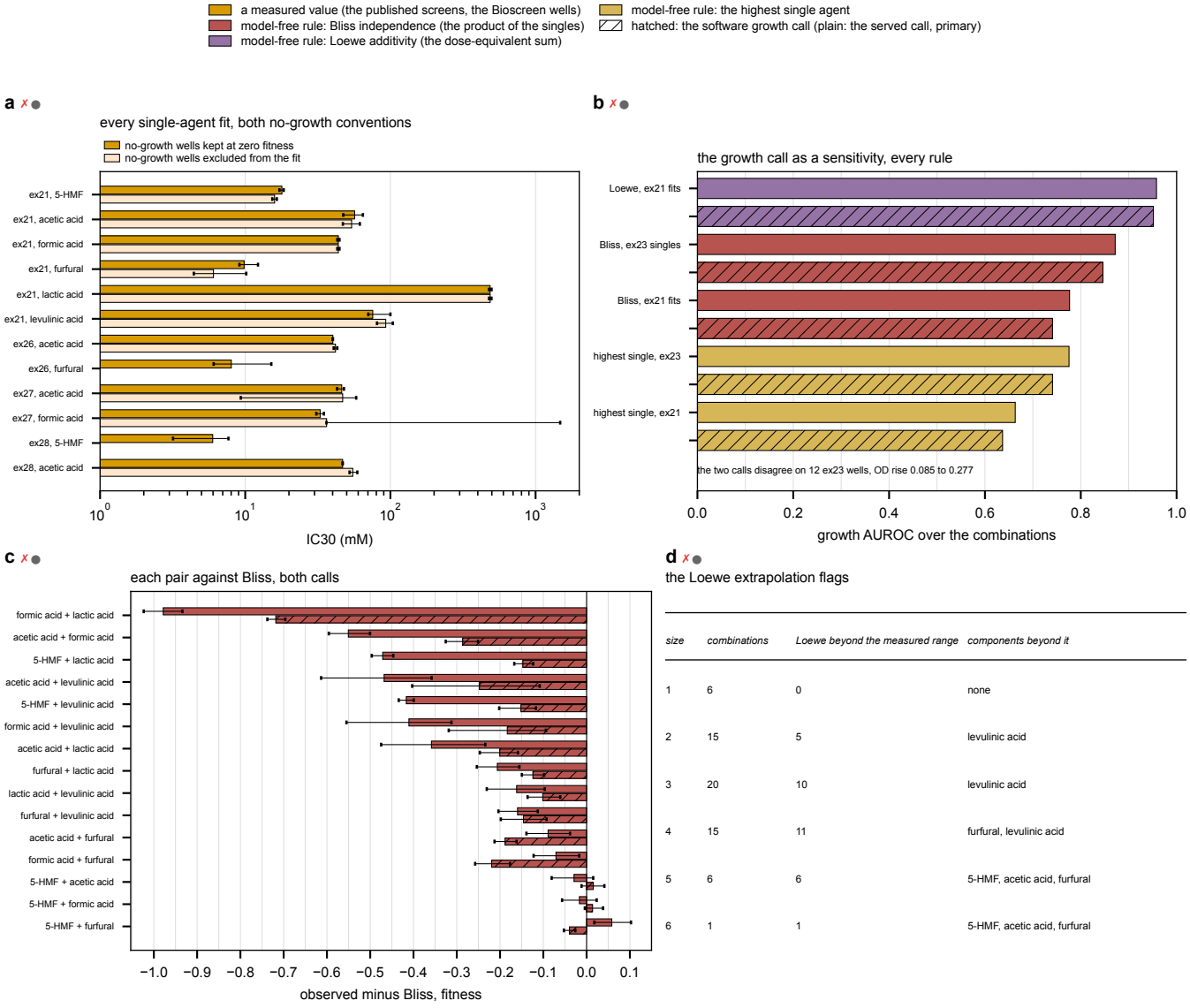


Figure 4: Supplement two: the single agents and the growth call. **a**, Every single-agent fit, 24 of them over the titration run and the three isobole axes, under both no-growth conventions: the darker bar keeps no-growth wells at zero fitness, the lighter one drops them from the fit. Intervals are the bootstrap. ✗● **b**, Every claim-one score under both growth calls, hatched for the software call. The two calls disagree on 12 wells of the combination run across 4 combinations, with optical-density rises from 0.085 to 0.277. ✗● **c**, Each pair against Bliss with its replicate-bootstrap interval, under both calls. Under the served call 12 of 15 pairs are synergistic, 2 additive and 1 antagonistic. ✗● **d**, Where the Loewe rule had to extrapolate past the measured dose range: 33 of the combinations, and 0 of the isobole fronts fell outside their grid. ✗●

Sources for fig. 4	
a	`single_agent_fits.csv`, `single_agent_bootstrap.csv`. Status: on file.
b	`mixture_scores.csv` (both calls), `wetlab_growth_call_check.csv`. Status: on file.
c	`pair_deviation.csv`. Status: on file.
d	`mixture_combinations.csv` (the extrapolation flags), `isobole_summary.csv`. Status: on file.

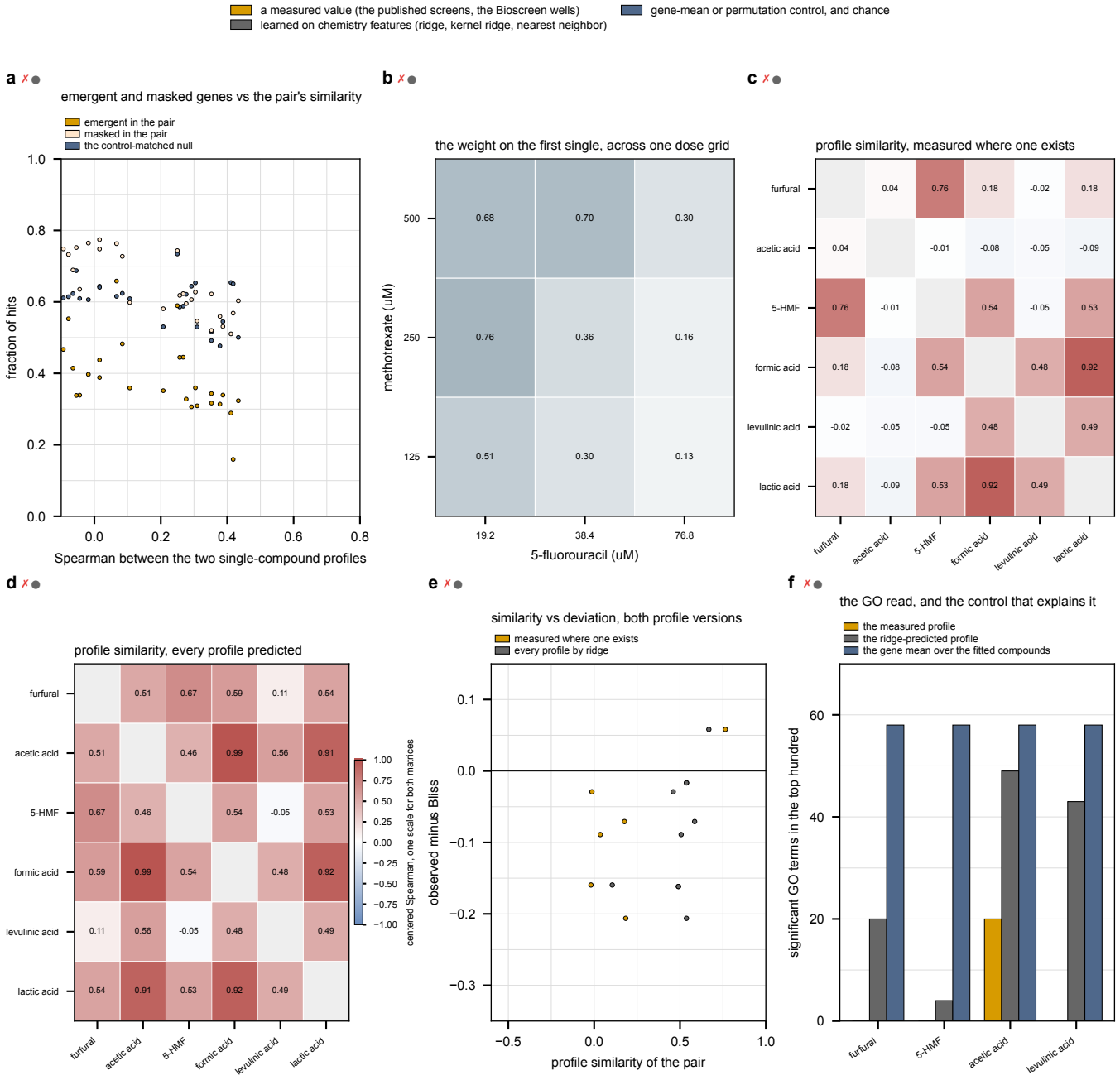


Figure 5: Supplement three: how profiles compose, and what the predicted ones carry. **a**, Emergent and masked gene rates of each published pair against how alike its two singles are, with the control-matched null beside them. The median emergent fraction is 0.355 against a null of 0.621, and the masked fraction 0.649; the only two near-replicate singles in the screen disagree on 0.536 to 0.603 of their own hits, which is the noise floor these rates sit inside. ✗● **b**, The one dose grid the published screen holds, methotrexate against fluorouracil: the weight the free linear fit puts on the first single moves from 0.128 to 0.757 across 9 dose pairs, while the mean wins at 9 of them. ✗● **c**, **d**, Profile similarity among the six inhibitors, measured where a measurement exists and then with every profile predicted. Measured, the strongest pair reads 0.924; predicted, 0.986, because the small acids have near-identical predicted profiles. One scale carries both panels. ✗● **e**, Similarity against the pair's departure from Bliss, both versions: 0.204 at a permutation p of 0.462 with measurements in, and -0.225 at 0.425 with everything predicted. ✗● **f**, The GO read: significant terms in the top hundred sensitive genes of the measured profile, of the ridge-predicted profile, and of the gene mean over the fitted compounds. The control carries 58 terms and the predicted profiles share 48 of them, which is what the predicted enrichment is. ✗●

Sources for fig. 5	
a	`het_emergent_masked.csv`, `het_near_replicates.csv`, `het_summary.json`. Status: on file.
b	`het_rule_fit.csv`, the methotrexate and fluorouracil family. Status: on file.
c, d	`profile_similarity.csv`, centered target, both versions. Status: on file.
e	`similarity_vs_deviation_pairs.csv`, `similarity_vs_deviation.csv`. Status: on file.
f	`go_overlap_measured_vs_predicted.csv`, `nominations_summary.csv`. Status: on file.

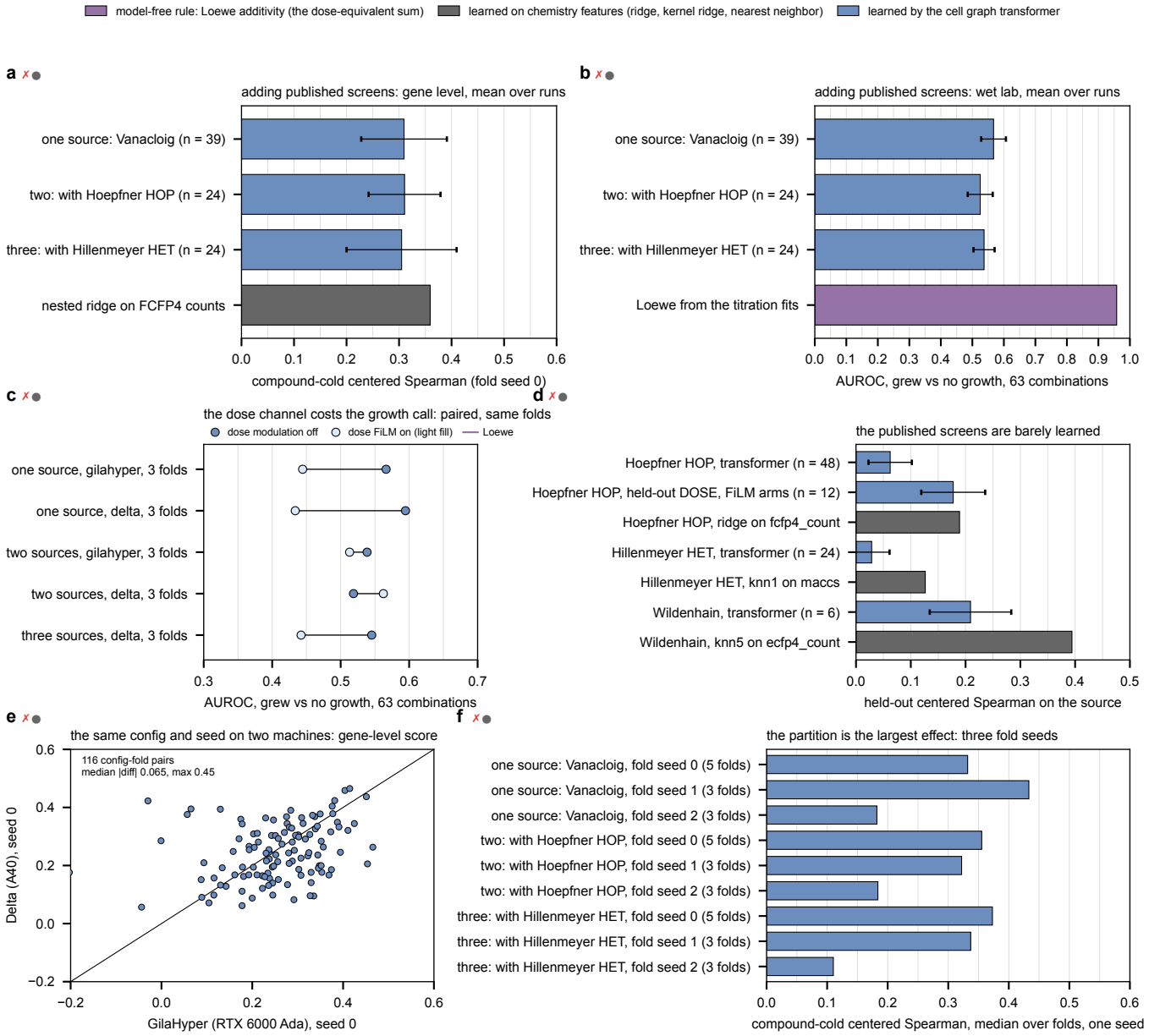


Figure 6: Supplement four: what the mixture rounds say, read over both machines. Rounds three and four of the mixture trainer, every run on fold seed 0 unless a panel says otherwise; a run is one seed on one fold on one machine, and an arm run on both machines contributes both copies. **a**, The one-, two- and three-source arms on the published side, mean over runs with its standard deviation, 39, 24 and 24 runs: 0.310, 0.311 and 0.305, against nested ridge at 0.359. The run spread is 0.082 to 0.105. ✗● **b**, The same runs on the wet lab, growth AUROC over the 63 combinations under the served call: 0.567, 0.525 and 0.537, against Loewe at 0.958. These are the gene readout's arms; the same ladder under the Bliss cell readout, at three seeds, is fig. 10. ✗● **c**, The dose channel, paired: the same arm with the log-molar dose modulation on and off, same machine, same folds, one seed; 5 pairs, FiLM lower on 4 of them, and the furfural-grid residual more negative on 5 of them. ✗● **d**, What the transformer predicts on the published screens it learns from: the held-out compound score per source, mean over the plain arms that carry it, beside the chemistry-feature learner's compound-cold score on the same source (Hoepfner 0.062 against 0.189, Hillenmeyer 0.029 against 0.126, Wildenhain 0.209 against 0.394); and the held-out DOSE of a seen Hoepfner compound under FiLM, 0.177. ✗● **e**, The same config at the same seed on the two machines, every round, 116 config-fold pairs: median absolute difference 0.065, largest 0.452, Spearman between the copies 0.332. That is the run-to-run floor at a fixed seed. ✗● **f**, The partition: each source-set arm on three fold seeds, one seed, median over the folds on file. The single-source arm reads 0.332, 0.433 and 0.182. ✗●

Sources for fig. 6

a, b	every `results/mixture/r3*/*_scores.csv` and `r4*/*_scores.csv` on GilaHyper and `results/mixture-delta/r3*/*_scores.csv` pulled from Delta, seed members of the arms `mix_van_e150`, `mix_vanhop_e150` and `mix_vanhophet_e150`; the ridge bar from the 038 ladder, the Loewe bar from `mixture_scores.csv`. Status: on file.
c	the same files, the `_film` arm against its dose-off twin on the folds both have, seed 0. Status: on file.
d	the same files, rows whose task is the auxiliary source (`heldout_compound`, `heldout_dose`); the chemistry-feature bars from `source_basis.csv`. Status: on file.
e	the same files, every config and fold scored on both machines at seed 0. Status: on file.
f	the same files, the `_fs1` and `_fs2` arms. Status: on file.

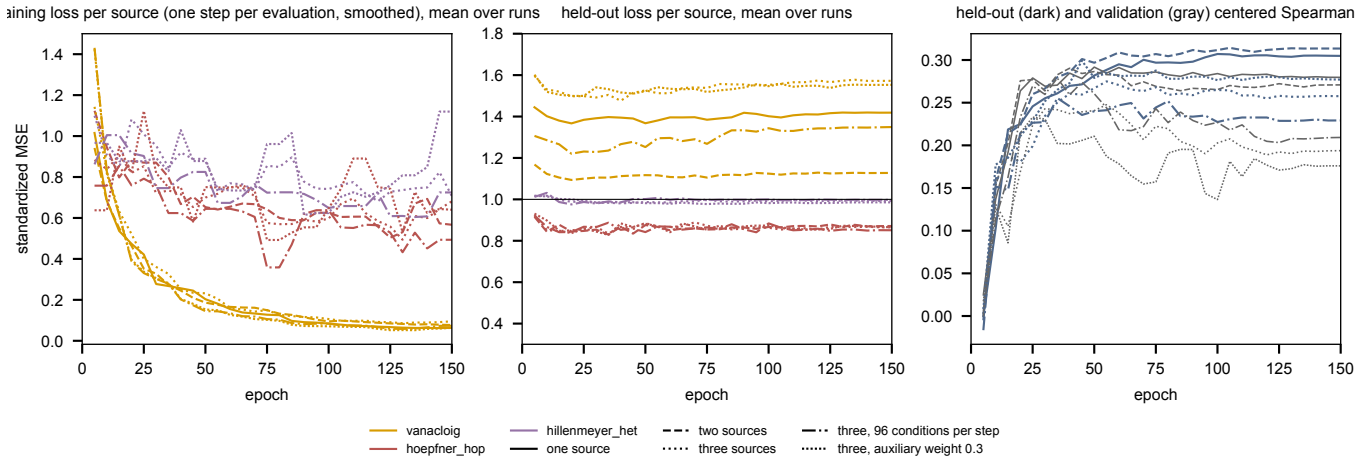


Figure 7: Supplement five: is the trainer converged, per source. Mean curves over every round-3 and round-4 run on both machines of the one-, two- and three-source arms and two three-source variants; the learning rate is zero at epoch 150 on every arm. **a**, The training loss per source, one step per evaluation, smoothed. The target's falls 0.545 to 0.062; Hoepfner's 0.865 to 0.573; Hillenmeyer's RISES, 0.869 to 0.824. ✗● **b**, The held-out loss per source. The target's removes 0.144 of its first value, bottoms near epoch 54 and rises 0.125 by the end; Hoepfner's removes 0.102; Hillenmeyer's 0.043; Wildenhain's 0.449, bottoming near epoch 100 and still falling at 150. ✗● **c**, The held-out and validation centered Spearman by epoch: the held-out score plateaus near epoch 60 to 75 while the held-out loss has already turned, so the ranking holds as the scale drifts. ✗●

Sources for fig. 7	
a to c	every `results/mixture/r3*/*_history.csv` and `r4*` on GilaHyper and `results/mixture-delta/r3*/*_history.csv` from Delta, read by `scripts/r3_convergence.py` into `results/r3_convergence_runs.csv` and `r3_convergence_arms.csv`. Status: on file.

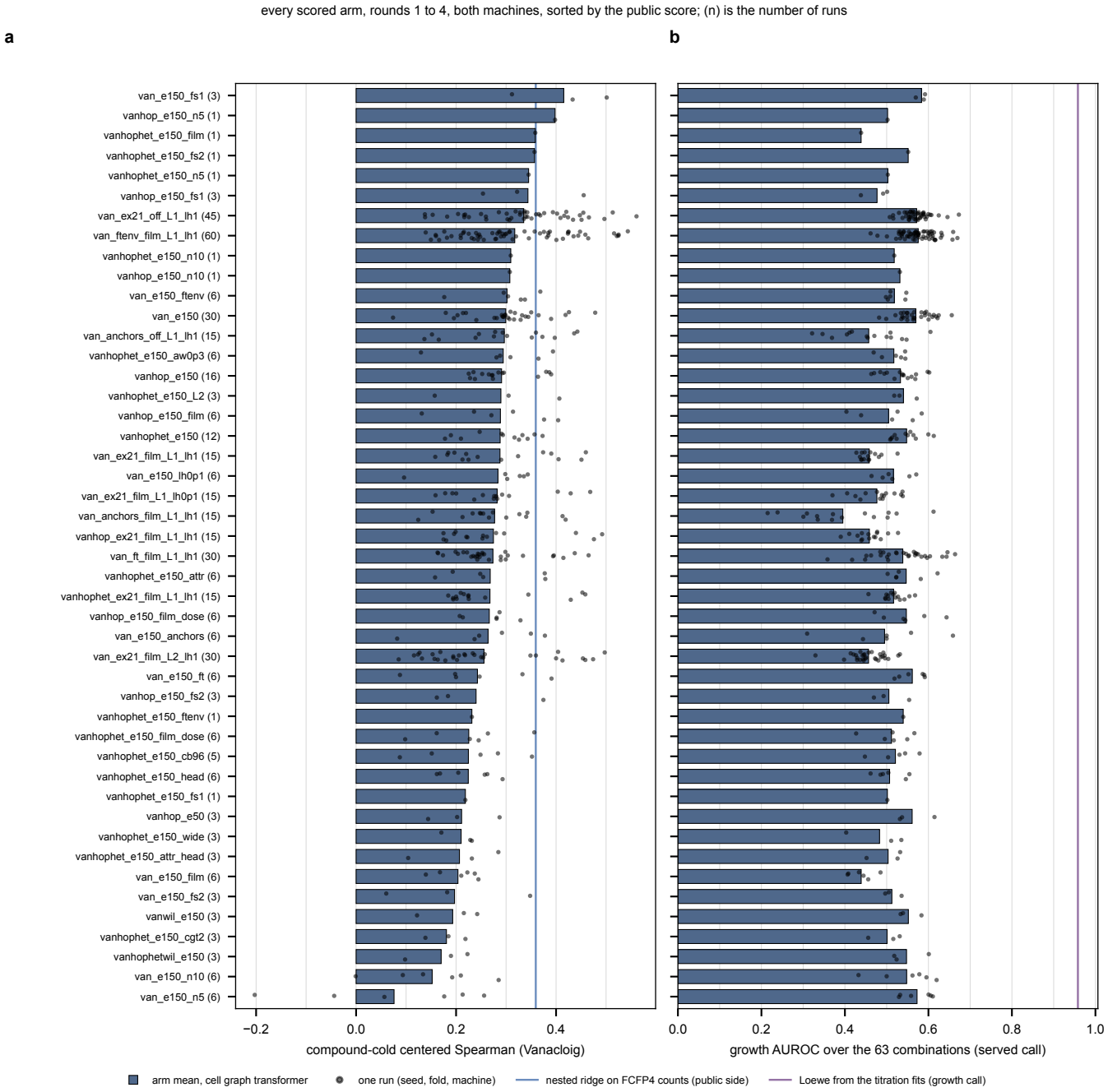


Figure 8: Supplement six: every scored arm of rounds one to four, both machines. One row per arm, every run a point (one seed on one fold on one machine) and the arm's mean a bar, sorted by the published score; rounds one and two (pool-and-last, 50 epochs) sit on the same axis as rounds three and four (the held-out protocol, 150 epochs). **a**, The compound-cold centered Spearman on Vanacloig against nested ridge at 0.359. **b**, The growth call over the 63 combinations against Loewe at 0.958. The spread of the points within an arm against the range of the arm means is the noise floor: no arm clears ridge and none approaches Loewe. ✗●

Sources for fig. 8	
a, b	every `*_scores.csv` under `results/mixture/` and `results/mixture-delta/`, drawn by `scripts/r3_plots.py`; the reference values from `results/figure_gate_numbers.json`. Status: on file.

2.1 The encoders, and how they were compared

Every encoder in this document was scored on ONE basis, so the sheet's bars are comparable: a compound-cold split of the Vanacloig profiles (the compound is held out with every one of its doses; 32 compounds on each of three fold seeds, 96 compound-evaluations), the centered Spearman across strains between the predicted and the measured profile of the held-out compound, and the median over the evaluations as the score. Three kinds of encoder sit on that basis.

- **Fixed chemistry encoders with a shallow learner** (the ladder, 038): 15 ways of embedding a compound, four fingerprint families (FCFP4 and ECFP4 as counts or bits, MACCS keys), RDKit's 2D descriptors, seven pretrained molecular models (mol2vec, MolE, Uni-Mol, ChemBERTa-2 in both pretraining modes, MoLFormer-XL, a ZINC RoBERTa) and three concatenations, each run with kernel ridge and nearest neighbors under linear, RBF and Tanimoto kernels, 62 learner-by-encoder arms, every regularization chosen by a nested fold. The best is nested ridge on FCFP4 counts at 0.359; the best pretrained model is Uni-Mol v1 (3D) at 0.323; the lowest single encoder is ChemBERTa-2 MLM at 0.219.
- **Learned environment encoders** (038 rounds 9 and 10): the bilinear control (a compound embedding read against a learned gene embedding, no attention) and the environment encoder (the compound token with its dose in front of the gene tokens of the cell graph transformer), trained on the same folds, three seeds each; the encoder is read against its control paired on compound and fold, median difference -0.036, up on 37 of 96.
- **The mixture trainer** (040), the same transformer with the published screens and the host records in one objective, which scores on its own basis (the 28 compounds with a molar dose, folds 0 to 2): 0.309 for the gene readout and, under the cell readout, inside its fold noise (section 4).

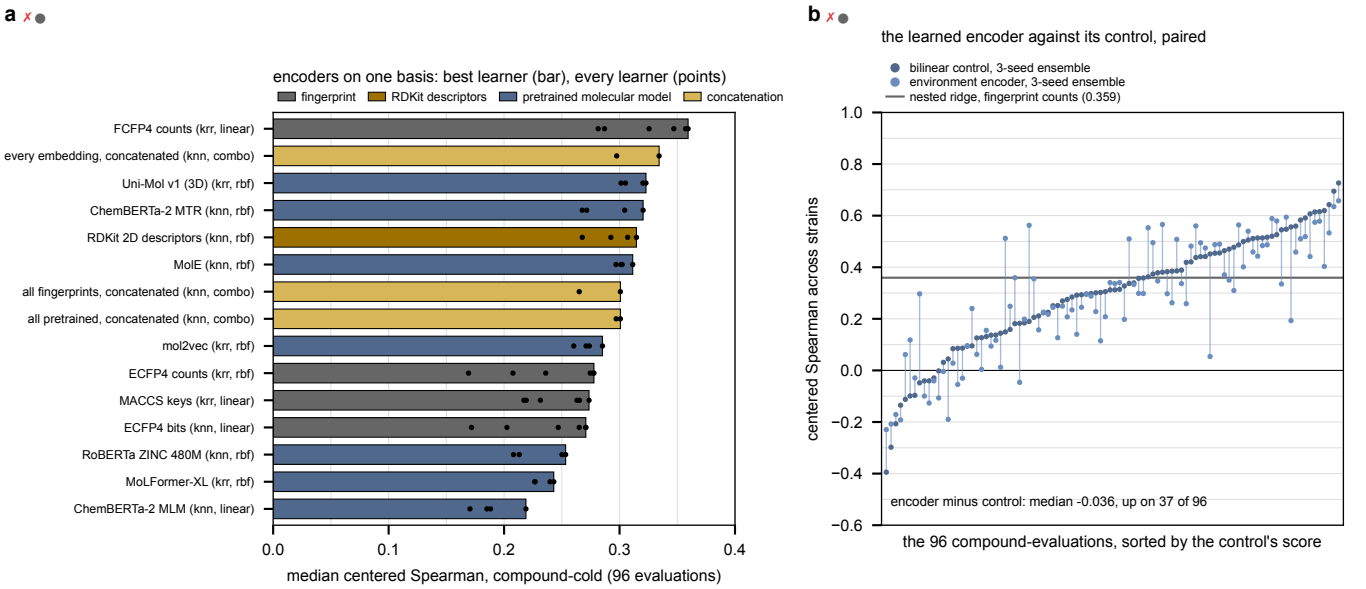


Figure 9: Supplement seven: the encoders compared. a, Every chemistry encoder of the ladder on the one basis, its best learner as the bar (the learner named in the label) and every learner it was run with as a point; color is the encoder family: fingerprints, descriptors, pretrained models, concatenations. Nested ridge on fingerprint counts leads; no pretrained model reaches it. **b**, The learned environment encoder against its bilinear control, paired on compound and fold over the 96 evaluations, sorted by the control's score; the line is nested ridge. The encoder is not above its control (median difference -0.036) and neither is above ridge. ✗●

Sources for fig. 9

a	`038/ladder/ladder_r2_summary.csv`, the centered target at the full compound count, every learner-by-encoder row. Status: on file.
b	`038/factorized/r9_control/*_scores.csv` and `038/factorized/r10_envenc/*_scores.csv`, the three-seed ensemble per fold, merged on compound, fold and fold seed. Status: on file.

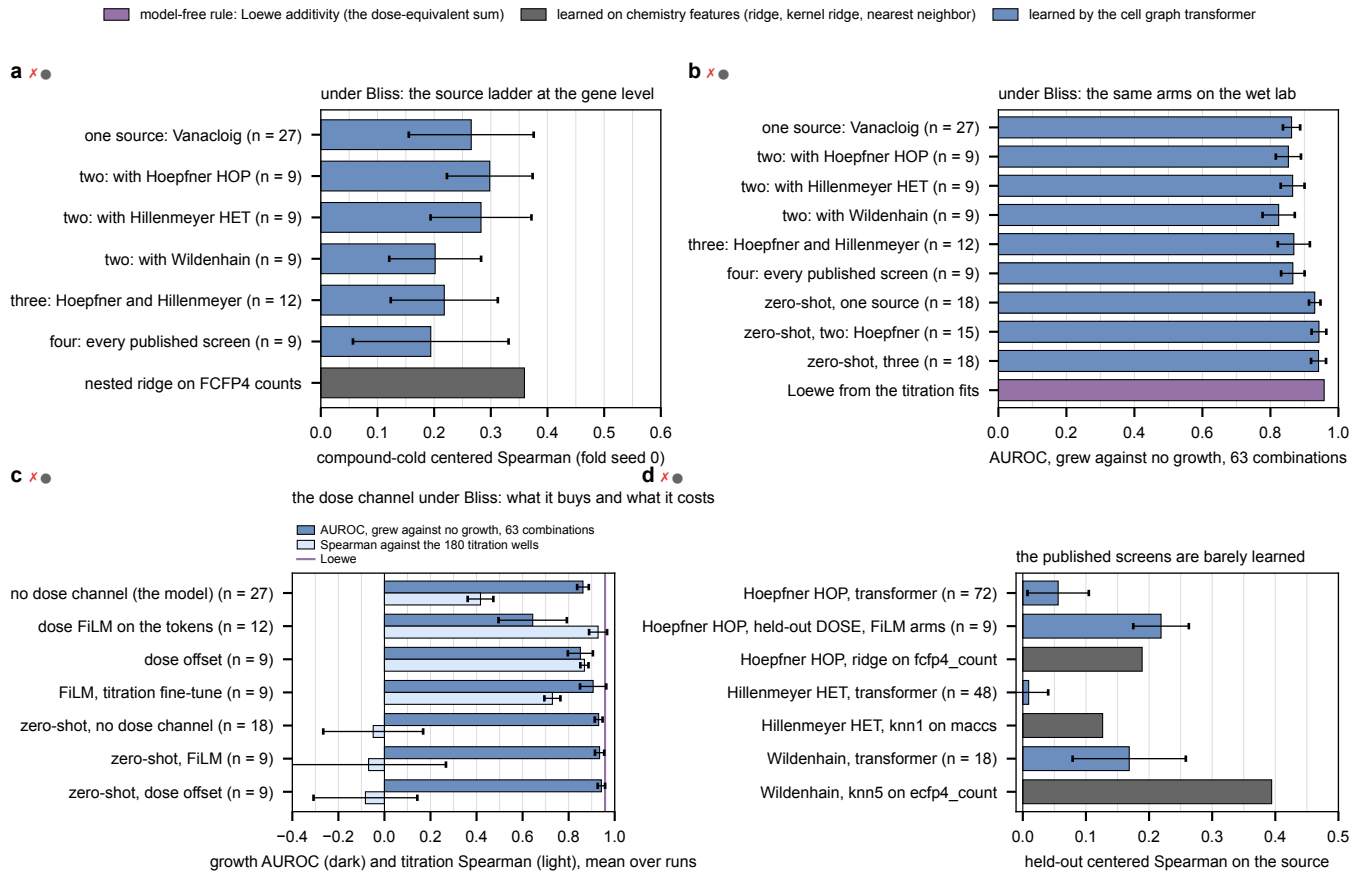


Figure 10: Supplement eight: round ten, the Bliss cell readout at three seeds. Every arm is three seeds on the three screening folds of fold seed 0, read on both machines where it ran on both; the bar is the mean over runs, the whisker the standard deviation. **a**, The source ladder on the published side: one source 0.265, with Hoepfner 0.298, with Hillenmeyer 0.283, with Wildenhain 0.202, three 0.218, four 0.194, run spreads 0.110 to 0.137, against nested ridge at 0.359. No published screen lifts the target, and the order is more sources, lower, inside one spread. ✗● **b**, The same arms on the wet lab, growth AUROC over the 63 combinations: the fitted ladder reads 0.862 to 0.866 and does not move with sources; the zero-shot ladder (anchors only, never a titration well) reads 0.930, 0.943 and 0.942 on one, two and three sources, every value above the Bliss rule on the measured singles (0.872) and the last at Loewe's (0.958). Three runs per arm; the steps are inside one spread, so this is a measured ordering rather than a cleared difference. ✗● **c**, The dose channel: each arm's growth AUROC (dark) beside the Spearman of its host prediction against the 180 titration wells (light), a fit for the arms trained on the titration and a prediction for the zero-shot arms. FiLM on the tokens reads the titration at 0.928 and composes at 0.644; the dose offset reads it at 0.869 and composes at 0.851, against 0.862 with no channel; the titration fine-tune with FiLM reads 0.729 and composes at 0.906. Zero-shot, no channel reads the titration, and the dose offset composes at 0.942. ✗● **d**, What the Bliss arms predict on the published screens they learn from, the held-out compound score per source over every round-ten arm that carries it: Hoepfner 0.056 against 0.189, Hillenmeyer 0.009 against 0.126, Wildenhain 0.169 against 0.394; the held-out Hoepfner dose under FiLM, 0.219. The screens are not learned at a held-out compound, so a screen could not have helped the target at one. ✗●

Sources for fig. 10

a, b	`results/mixture-delta/r10*/*_scores.csv` pulled from Delta and `results/mixture/r9*/*_scores.csv` on GilaHyper (the seed-0 copies of the one- and three-source Bliss arms), seed members on fold seed 0; the ridge bar from the 038 ladder, the Loewe bar from `mixture_scores.csv`. Status: on file.
c	the same files, the `ex23` growth AUROC and the `ex21` all-wells fitness Spearman of the seven dose arms. Status: on file.
d	the same files, rows whose task is the auxiliary source (`heldout_compound`; `heldout_dose` of `mix_vanhop_e150_cell_bliss_film_dose`); the chemistry-feature bars from `source_basis.csv`. Status: on file.

3 What each panel still needs ✗

Table 1: What each panel still needs before it is real. The status column repeats the panel's mark in words: on file (filled circle), every drawn value is in a committed result file; partial (half circle), a run is unfinished or a planned arm has no file; idea only (open circle), a layout or nothing run. No panel is approved; approval is the section status and is the author's.

panel		what remains
concept; main, a	○	Redraw as one draw.io composition at print size. The counts in it are on file; the boxes and arrows are a layout, and the mixture path they describe (a set of compound tokens, each modulated by its own log-molar dose) is the proposed model and has not run.
main, c	○	Same: a layout whose numbers are on file. It merges into the concept sheet if the figure is cut from four rows to three.
main, e; SI-1, a	◗	The second factorized round is producing. Two folds of the bilinear control and two of the environment encoder were on disk when the sheet was drawn; the round's remaining folds, the nine-seed ensembles and the stack with nested ridge are not written yet. Round eleven on Delta adds the width and depth axis.
main, f	◗	The encoder arm of the learning curve has to be rerun on the corrected store; only the ridge arm has a file. The planned round also adds the other pooled sources to the gene-level task, which is the second line this panel would carry.
main, i	◗	The transformer's compound-cold profiles for furfural and 5-HMF come out of the same running round. Until then the panel shows ridge against the measured profile and the screen's own repeat, and the transformer row is a label with no bar.
main, j	◗	The host head's predicted single-agent curves now exist for the anchors-only arm (zero-shot) and the arms that train on the titration run (section 5); the panel is not yet drawn from them.
main, k, l	◗	Round-one model predictions over the combinations and the isobole grids are on file for 5 arms and below the rules on every readout (table 2); the boundary scorer of section 5 has not yet been run on the model surfaces. The panels ship with the rules and the model side by side and the text says so, unless a queued arm clears a bar.
boundary, a–f; rule choice, a, b	●	Nothing. Both figures of section 5 are drawn by committed scripts from committed result files. Open: whether the boundary overlap replaces growth AUROC as the primary wet-lab statistic.
main, b, d, g, h	●	Nothing. Every drawn value is in a committed result file. These four are what the figure can claim today.
SI-1, b, c, d	●	Nothing.
SI-2, all	●	Nothing.
SI-3, all	●	Nothing, with one caveat the caption carries: the similarity matrices mix a measured profile for two inhibitors with predicted profiles for four, and the all-predicted version exists so that mixture is visible rather than hidden.

The running and planned rounds, and what each one fills. The second factorized round of the corrected-store campaign is on the two GilaHyper cards named for it, one job for the bilinear control and one for the environment encoder; it fills main **e**, main **j** and the first column of SI-1 **a**. Round eleven of the same campaign goes to Delta under the account the plan names, never a preemptible partition, and widens the ladder rather than deepening it. The first mixture round of this experiment is the one that fills the bottom row: about ten arms by three seeds, crossing the pooled sources against the host-training recipe (the published dose anchors alone, the anchors plus the titration run, and a fine-tune on the titration run) against the dose treatment. It starts once the mixture trainer passes its processor smoke test, and its launchers are the pair of scripts the experiment folder already carries.

The cut from four rows to three. The experiment note proposes merging rows two and three. The merge that costs least is to move main **c** into the concept sheet, where the same boxes already appear, and to put main **f** and main **h** side by side as one panel on two axes, since both are about what more data buys: more compounds on one axis, a second medium on the other. That leaves nine panels: the task, the data, the ladder, the transformer against ridge, the merged data-and-dose panel, the composition rule, the predicted profile, the single agents, and the combinations with the isoboles as its inset. The alternative, dropping the single-agent panel, is worse: it is what every rule in the bottom row is computed from, and a reader who cannot see the Hill slopes cannot judge why Bliss and Loewe disagree as far as they do.

4 State on the day the sheets were drawn ✗

What had finished. The processor round of this experiment: the four pre-registered claims scored, which is every panel marked on file above. The ladder of the corrected-store campaign, giving main **d** and SI-1 **a**. The learning curve of the earlier campaign, giving main **f** in its ridge arm. The pooled table and its axes, giving main **b**.

What finished while the sheets were drawn. The second factorized round of the corrected-store campaign, submitted in the small hours of the same day on two GilaHyper cards: one job for the bilinear control, one for the environment encoder, each fifteen configurations of five compound-cold folds on three fold seeds, three seeds per fold. Both completed, giving main **e** in full: the control's three-seed ensemble at 0.312 and the encoder's at 0.297 over 96 compound-evaluations, against nested ridge at 0.359, with the encoder -0.036 against its own control when the two are paired on compound and fold. Neither head reaches the fingerprint ridge on this store, which reverses the earlier campaign's reading and is the result the two Delta rounds below are meant to explain or confirm. The sheet re-reads the directory every time it is drawn, so a later round replaces these values rather than adding to them.

What had not started. The first mixture round, which is every panel of the bottom row in its model arm. The trainer and its configuration groups are owned by the implementation thread and are not part of this document.

What to expect, labeled. Hypothesis (untested): the environment encoder ties nested ridge on the corrected store, as it did on the earlier store, so main **e** closes with the two within a hundredth of each other and the figure's published-benchmark row says that architecture is not where the gain is. Hypothesis (untested): the encoder's compound-cold profiles for the two inhibitors in store read no better than ridge, so main **j** keeps its shape and the predicted profiles stay generic. Hypothesis (untested): a dose-aware host head fitted on the titration run beats the highest single agent on growth, which reads 0.663, and does not beat Loewe, which reads 0.958, because Loewe already uses the fitted single-agent curves and the model has no information about this host that those curves do not carry. If that last one holds the honest figure is the model-free one, and the bottom row says so in the caption rather than in a footnote.

What would change the figure's claim. A model arm that beats 0.958 on growth, or 0.573 on fitness over the combinations that grew, moves the bottom row from a validation of arithmetic to a result about transfer. A model arm that does neither still earns the row, because the comparison is the point: the rules are computed from six titrations on the same strain, and a model trained on 5,463 published compounds that cannot beat them is the measurement the section needs to report.

The same day, evening. The first mixture round ran: two arms on the GilaHyper cards named for this work, two on Delta, two more on GilaHyper by the evening, and the rest on Delta overnight; table 2 holds the arms with files here. The third hypothesis above held in its pessimistic half: no arm beats Loewe on growth or Bliss on fitness, and the dose-aware arms do not beat the highest single agent either. The readout was then reframed around the alive and dead boundary (section 5), which is model-free and splits by pair, and the leave-one-pair-out test of rule-choice predictors was added as the bar a learned model must clear. Round two began with the replication of the dose-off joint arm on two further fold seeds.

Three days on, 2026.10.15. Ten rounds are on file, every arm on both machines. The readout is the cell, the composition is Bliss, the model sits at the rules' level zero-shot on the private combinations (0.930 with one source, 0.942 with three) and the dose offset gives it a dose axis at no published cost; the published bar is still ridge's. The bottom row ships with the model beside the rules, which was the third open decision. What is not on file: a deviation term beyond Bliss, a gene-cold split on the published screens, and any arm above ridge on the published side.

5 The boundary readout, and what the first model arms say ✗

For several inhibitors the useful prediction is the shape in dose space inside which the host still grows, the line between alive and dead at each concentration. The fixed-dose combination plate gives one point per combination, so the boundary lives in the three isobole grids and in the single-agent curves that fix where it meets the axes. Bliss multiplies surviving fractions, two independent hits; Loewe adds doses as fractions of each compound's own dose for the same effect, one shared capacity. With the steep single-agent curves of this host the two fronts sit far apart, so which one a pair follows is a large, visible difference, and it is scored here directly: a rule's alive region is the interior cells where its prediction clears the grid's growth threshold, compared with the observed alive region by the overlap of the two sets (the Jaccard index), the cells wrong in each direction, and the growth edge, the highest dose called alive in each column, in grid steps.

Post-processing the observed region. A dose-response boundary is monotone: more inhibitor never rescues growth. A grown cell sitting above a dead one is therefore a call error, and the observed region is closed downward before it is scored (a cell stays alive only if every cell with lower doses on both axes is alive). The furfural and formic acid grids are already monotone. The 5-HMF grid, flagged since the processor round for its growth islands, loses 4 interior cells to the closure, going from 17 alive cells to 13; the figure marks each dropped cell with a cross. Any predicted surface gets the same closure before it is scored as a boundary.

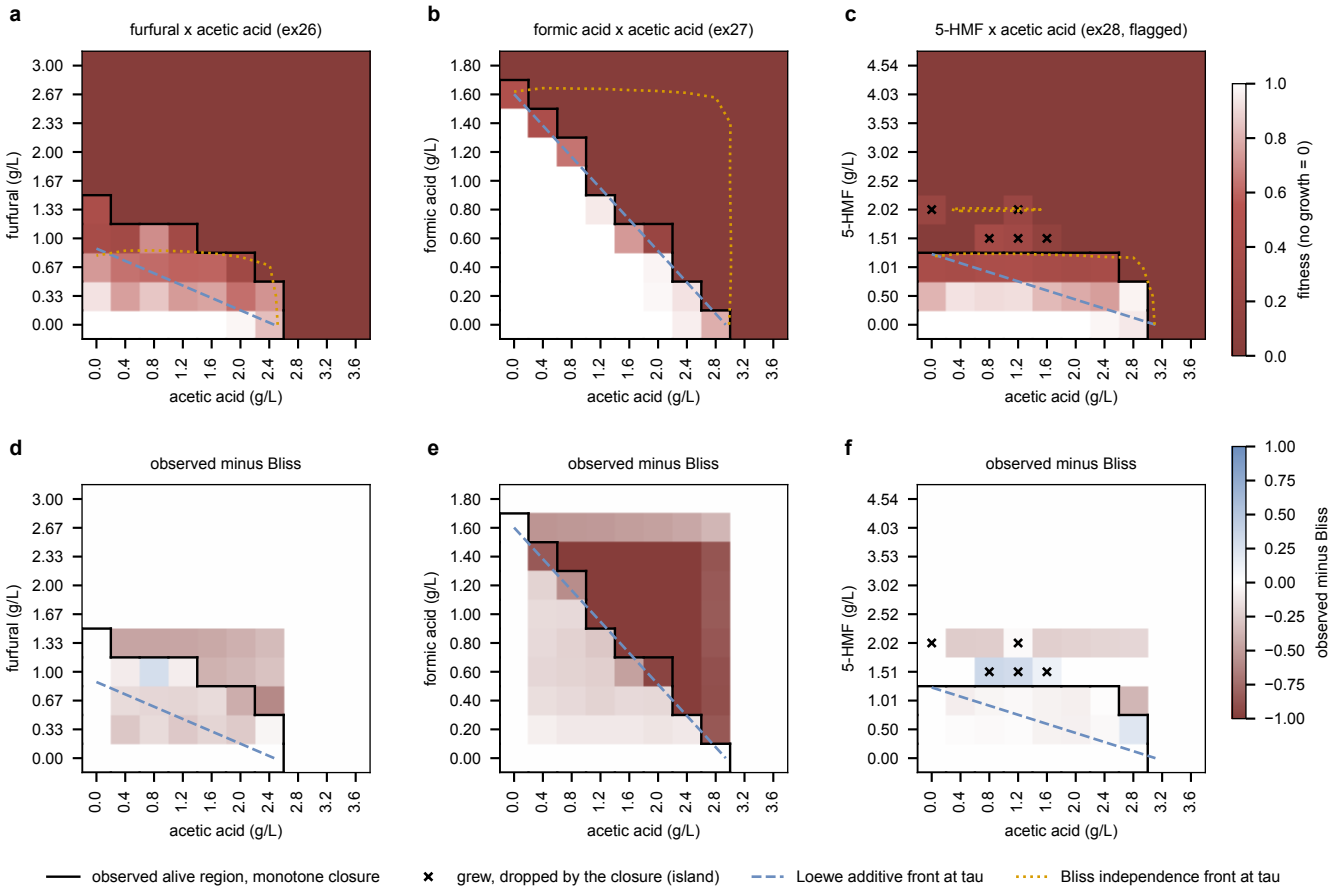


Figure 11: No single rule draws the boundary. **a** to **c**, Observed fitness over each isobole grid (no growth drawn as zero), the black outline around the monotone closure of the cells that grew, a cross on each grown cell the closure drops, and the two fronts at the grid's growth threshold: Loewe additivity dashed in blue, Bliss independence dotted in orange. Furfural with acetic acid follows Bliss, overlap 0.733 against 0.286 for Loewe, which calls 10 grown cells dead. Formic with acetic acid follows Loewe, overlap 0.875 against 0.436 for Bliss, which keeps 31 dead cells alive. 5-HMF with acetic acid follows Bliss, overlap 0.765 against 0.385. The growth edge along acetic acid tracks the winning rule within 0.444 (furfural), 0.333 (formic acid) and 0.778 (5-HMF) steps on average. ✗● **d** to **f**, Observed minus the empirical Bliss surface on a symmetric scale, with the Loewe front dashed. ✗●

Sources for fig. 11

a–c	`isobole_cells.csv` (observed, Bliss and Loewe per cell, the raw and the monotone growth call), `isobole_boundary.csv` (the overlap, the cells wrong each way and the edge error per grid, rule and observed region), `single_agent_fits.csv` (the axis Hill fits the Loewe front is drawn from). Served growth call; the isobole runs carry no software call. Status: on file.
d–f	`isobole_cells.csv`, `isobole_summary.csv` (the mean excess over Bliss and its bootstrap interval, printed in section 2). Status: on file.

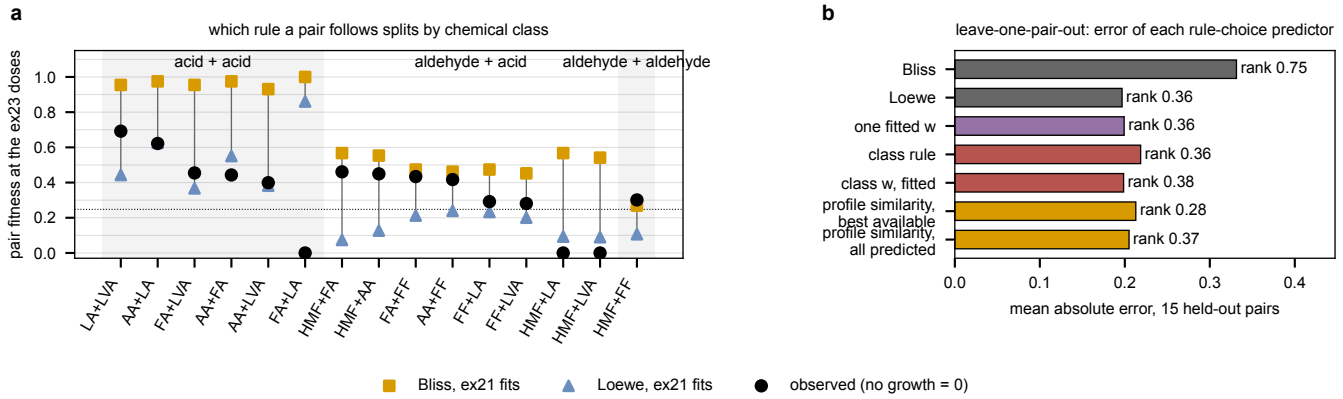


Figure 12: Which rule a pair follows, and whether anything predicts the choice. a, Each of the 15 fixed-dose pairs: observed fitness (no growth drawn as zero) against Bliss and Loewe from the titration fits, grouped by chemical class, the dotted line the growth threshold. All 6 acid pairs follow Loewe. The 8 aldehyde and acid pairs split 4 to 4, by the acid and not by the class: with acetic or formic acid the aldehyde follows Bliss, with lactic or levulinic acid it follows Loewe, and 5-HMF with either of those two grew in no well, which Loewe calls and Bliss does not. ✗● **b**, Leave-one-pair-out: each predictor gives a Loewe weight for the held-out pair, fitted on the other pairs, and the pair's fitness is that weight on Loewe plus its complement on Bliss. Mean absolute error over the held-out pairs, with the Spearman over the 12 grown pairs beside each bar. Pure Loewe has the lowest error, 0.197; pure Bliss the best ranking of the grown pairs, 0.748, because its error is a level bias that leaves their order intact. The growth call is not shown: with 12 of 15 pairs grown, Bliss scores 0.800 by predicting growth for every pair, a majority-class readout. A fitted global weight (0.199), the class rule (0.219), a fitted class weight (0.199) and profile similarity with the best available profiles (0.213) or the all-predicted ones (0.205) all land between the two rules. Nothing on file predicts the choice; with 15 pairs that is a small-sample statement. ✗●

Sources for fig. 12

a	`rule_choice_pairs.csv` from `mixture_combinations.csv` (served call) and `profile_similarity.csv`; the class of a pair is the chemical class of its two members. Status: on file.
b	`rule_choice_lopo.csv` and `rule_choice_lopo_pairs.csv`; the class hierarchy over all combinations (Loewe within a class, Bliss across) is in `rule_choice_higher_orders.csv`: growth AUROC 0.887 against Loewe's 0.958, with the smallest fitness error over the grown combinations (0.148 against 0.208). Status: on file.

What the figure now claims. The boundary is model-free and it is not one rule: Loewe for the acids, Bliss for the short acids with an aldehyde. The rule choice is the quantity a learned model would have to supply, and the leave-one-pair-out test is the bar it has to clear. That replaces "Loewe wins overall" as the statement of the bottom row.

The first model arms. The mixture trainer of this experiment ran its first round: the environment-encoder recipe of the corrected-store campaign with one compound token per inhibitor, a log-molar dose modulation (FiLM) that can be switched off, and a host head read at the mean over genes, fitted on the published dose anchors alone or on the anchors with the titration run, five compound-cold folds of one fold seed with three seeds each. The combination plate and the isobole grids are never trained on. 5 arms have score files here; two more, the dose-off pair, finished on Delta and await an rsync, and three are running there. Every arm sits below the model-free rules on every wet-lab readout.

Table 2: Round-one arms against the model-free bars, served growth call, median over the folds on file. The reference for growth is Loewe from the titration fits, AUROC 0.958; for the worst-scenario ranking over all combinations, Loewe, 0.772; for fitness over the combinations that grew, Bliss from the observed singles, 0.573; for the furfural grid, the mean of observed minus Bliss, -0.082; on the published side, nested ridge, 0.359. A negative grid residual means the arm predicts more growth than was seen.

arm	folds	growth AUROC	worst-scenario Spearman	fitness Spearman, grown	furfural grid, observed minus arm	published, centered Spearman
joint, anchors and titration, dose FiLM	5	0.475	-0.009	0.377	-0.751	0.261
anchors only, dose FiLM	5	0.295	-0.319	-0.346	-0.598	0.266
two-stage, head fine-tuned on the titration	5	0.556	0.093	0.216	-0.569	0.271
two-stage, head and environment encoder fine-tuned	5	0.582	0.154	0.188	-0.554	0.256
joint, two environment layers (still writing)	5	0.469	0.002	0.341	-0.765	0.252

What the arms say. The dose-aware joint arm fits the titration wells it trains on and then predicts the combinations at chance; the anchors-only arm, the zero-shot host, is below chance and ranks the grown combinations the wrong way. Switching the dose modulation off raises the growth AUROC in both host regimes (the Delta pair, recorded in the experiment note until its files arrive), which is the opposite of what a dose-aware model is for. Every arm's grid residual is negative: the arms predict more growth in the interior cells than was seen, and the dose-aware arms predict the most. Hypothesis (untested): the host head composes inhibitors near-additively or less, so each added inhibitor lowers predicted fitness by less than it lowers observed fitness, and the sharper the single-agent fit the larger that gap. On the published side every arm reads within a few hundredths of the earlier campaign's encoder and below ridge, so the wet-lab failure is not a model that was working elsewhere.

Where a learned model can still earn the panel. The rules need the single-agent curves. The regimes in which they have no prediction, or one they cannot improve with data, are the model's territory, and each needs its own baseline. Zero-shot singles: no Bioscreen well in training, the six titration curves predicted from published screens; the honest readout is the pooled ranking over all titration wells, since within one compound fitness falls with dose and any dose-aware model scores near perfectly there; on file the anchors arm reads weakly and the dose-off anchors arm reads nothing. One inhibitor unknown: five curves and the anchors known, the sixth withheld, the model predicts its curve and every combination that contains it; the baseline is the hybrid, the model's curve fitted as a Hill and fed to Loewe. Singles and pairs known, higher orders predicted: the fifteen pairs become training data and the forty-eight higher-order combinations the test; the baseline is a rule with one interaction parameter fitted on the pairs. Leave one pair out: the same, with the fifteenth pair as the test and the rule choice as the quantity learned.

Open decisions. First, whether the boundary readout, the overlap of alive regions and the edge error under the monotone closure, replaces growth AUROC as the primary wet-lab statistic of the bottom row; this document already scores both. Second, whether the fifteen pairs may become training data in the two regimes that need them; that breaks the pre-registered rule that the combination plate is never trained on, for those arms only, and the note would say so. Third, whether the bottom row ships with the model beside the rules, as the arm table shows it, or with the rules alone until an arm clears one of the bars. The first two are open; the third follows from whatever the queued rounds return.

Round three, read over both machines. The screen that followed round one (fig. 6) was read on 2026.10.12 with every arm scored on Delta and most of them on GilaHyper as well. Three readings. The same config at the same seed differs between the machines by a median of 0.065 and by up to 0.452 on the published score, so no single-seed difference in the screen is a finding. Over every run so far the one-, two- and three-source arms read 0.310, 0.311 and 0.305 on the published side and 0.567, 0.525 and 0.537 on the growth call: adding the published screens does not lift either readout, and the arms predict a cold compound on those screens at 0.062 (Hoepfner) and 0.029 (Hillenmeyer) against the chemistry-feature learners' 0.189 and 0.126. The dose modulation learns a held-out published dose (0.177) and lowers the growth call on 4 of 5 paired comparisons, which is round one's reading again under a protocol that selects on a held-out signal. Round four, six seeds on five folds of the three source-set arms, is what will turn the first reading into a measured null or a difference.

Rounds seven to ten, read over both machines. Round four returned the measured null: the three source-set arms at six seeds sit inside one spread on both readouts (fig. 6). Rounds seven to nine changed what a record is. Under the cell readout every record is a cell, the host the record with the empty genotype read through the shared head, and the published score did not move; what moved the private readout was the composition. With the compound tokens of a medium averaged, every arm read one fitness at one, two and three inhibitors (growth AUROC 0.48 to 0.57 over six runs); summed, the medium read fitter than its singles (0.27); read one inhibitor at a time and composed as Bliss independence, the same model orders the combinations at 0.862 fitted over 20 runs and 0.930 zero-shot, and its predicted fitness falls with the inhibitor count. Round ten (fig. 10) put three seeds on the questions. Sources: no published screen lifts the target, and the four-source arm is the lowest (0.265 to 0.194); the fitted wet-lab readout does not move with sources; the zero-shot readout does, 0.930, 0.943 and 0.942 on one, two and three sources, three runs each and the steps inside the spread. Dose: FiLM on the tokens reads the titration (0.928) and costs the host (0.644 against 0.862 with no channel), its published cost of the single-seed rounds being seed noise; the dose offset reads the titration at 0.869 at no published cost and composes at 0.851; the titration fine-tune with FiLM composes at 0.906.

What the published screens themselves can be predicted at is a narrower statement than it may read. The held-out compound score is the only thing measured on them: a compound the model never saw, its predicted profile across the HOP strains against the measured one, 0.056 over the round-ten arms against ridge's 0.189, and across the Hillenmeyer HET strains 0.009 against 0.126, the HET training loss rising rather than falling in every round. A seen compound at a held-out dose is a different question and is predicted (0.219). A held-out strain with every compound seen, the gene-cold split, has been run on Vanacloig alone, where it read near zero, and not on HOP or HET, so whether a new HOP or HET strain's response can be predicted is unmeasured rather than negative; the honest experiment is a gene-cold split on each screen scored as hit recovery over the strains called sick rather than a rank over every strain. Hoepfner's HIP arm is not in the pool at all, only its HOP arm and Hillenmeyer's HET arm. Hypothesis (untested): haploinsufficiency profiles are sparse, so a rank over all strains is a poor target for HET and a hit-level call would read differently.

6 Round eleven: the decoder takes the rules' functional form ✗

What the round asked. Ten rounds left three readings: composition is what buys the private readout, no published screen lifts the target or is learned at a held-out compound, and every dose arm over-predicts the isobole interiors. Round eleven tested five new parts of the model, each switchable and each defaulting to the round-ten arm, so every cell of the round is an ablation of one part. *Displacement*: the compound token moves every gene embedding along a learned direction, gated per gene, which is the environment's version of the encoder's deletion operator, against the round-ten design in which the compound tokens attend to the genes. *Cell-only*: the head reads the perturbed cell state and nothing else, so a compound's whole effect has to pass through the gene embeddings. *Hill form*: a second readout of the same state gives each inhibitor a dose-response curve, an IC50 and a slope, and the host's fitness is that curve at the medium's dose. *Composition*: the single-inhibitor readings of a mixture combine by Bliss independence, by Loewe additivity solved on the curves, or by the highest single agent, the three model-free rules of the figure now computed inside the decoder. *Pair term*: a learned correction on every pair of a mixture's compounds, zero at the start, which is the environment's analog of a genetic interaction. The 62 arms ran as 105 one-GPU jobs on Delta; 163 of their 186 configs are scored, the rest rerunning after Lustre write errors.

The control and the noise. Every arm is read against the round-ten Bliss arm, re-run in this round on the same folds, seeds and inputs. Its growth AUROC is 0.850 and its compound-cold Vanacloig score 0.265. A single seed scatters around its fold's three-seed ensemble by 0.111 on the Vanacloig score and 0.018 on the AUROC; the 63 combinations are the same wells on every fold, so the fold-to-fold spread of a host score, 0.041 for the control, is the floor a host difference has to clear. Everything here is one fold partition.

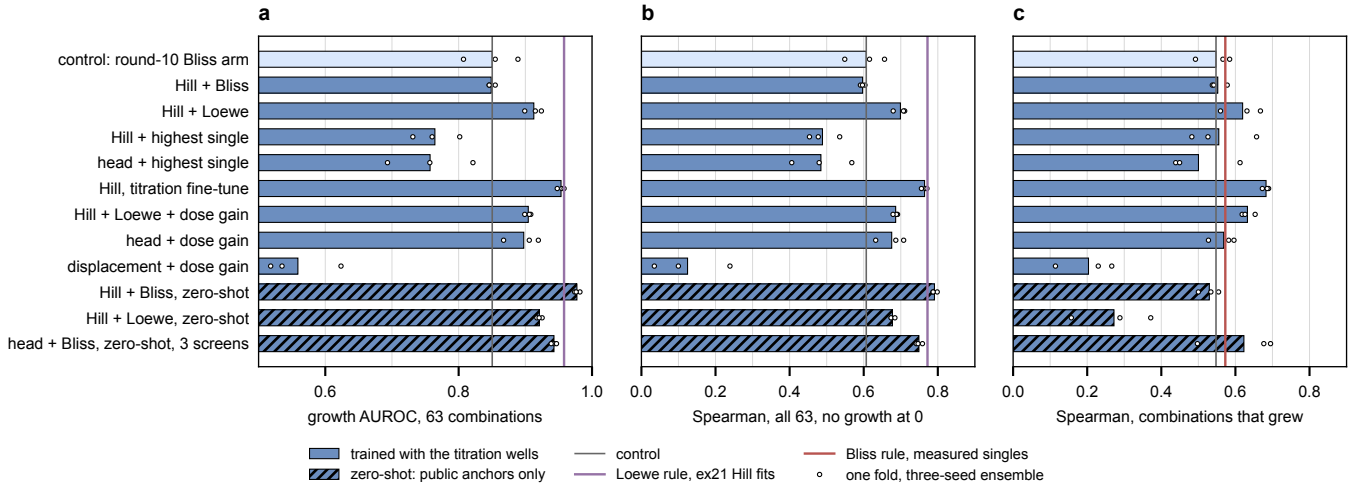


Figure 13: Round eleven on the wet-lab wells. One row per arm, the bar the mean over folds of each fold's three-seed ensemble, the open circles the folds. Hatched arms never saw a titration or a combination well: their only host records are the published IC30 anchors and the compound-free medium. Vertical lines: the control (gray), Loewe additivity from the `ex21` Hill fits (purple), Bliss independence from the measured singles (red). **a**, Growth AUROC over the 63 combinations. **b**, Spearman over the same combinations with no growth read as zero, the ranking a process engineer needs. **c**, Fitness Spearman over the combinations that grew. The displacement arm with the dose gain is the only arm that falls far below the control.

The reading. The Hill form composed by Bliss and trained on the published anchors alone orders the combinations at a growth AUROC of 0.977 (folds 0.974 to 0.982) against Loewe's 0.958, and ranks all of them at a worst-scenario Spearman of 0.791; it is the first arm of the experiment above the Loewe rule, on one partition and three seeds, with single seeds from 0.942 to 0.982. Fitted on the titration, the decoder's rules order as the model-free rules do: Loewe 0.913, Bliss 0.849, the highest single agent 0.764. The Hill fine-tune is the balanced arm (0.954 growth, 0.683 on the combinations that grew, against Bliss's 0.573). Displacing the gene embeddings leaves the published score (0.276 to 0.325) and the host unchanged, and with the dose gain it reads the titration at 0.937 while composition collapses to 0.559. No arm reaches nested ridge's 0.359 on the published side, no dose channel beats none on a held-out published dose (0.258 without one), and the gene-cold split reads zero (-0.001 to 0.007). Figure 14 wires the leading arm.

Table 3: Round eleven on the wet-lab wells. Fold ensembles of three seeds, averaged over the folds named, fold seed zero, served growth call. Growth AUROC and the worst-scenario Spearman are over the 63 combinations (no growth read as zero for the second); grown is the fitness Spearman over the combinations that grew; titration is the Spearman over all `ex21` wells, a train fit except for the zero-shot arms, which never saw one.

arm	folds	growth AUROC	worst-scenario Spearman	grown Spearman	titration Spearman
Loewe rule, ex21 Hill fits	–	0.958	0.772	–	–
Bliss rule, measured singles	–	–	–	0.573	–
control: round-10 Bliss arm	3	0.850	0.607	0.548	0.437
Hill + Bliss	3	0.849	0.597	0.552	0.758
Hill + Loewe	3	0.913	0.699	0.619	0.763
Hill + highest single	3	0.764	0.489	0.555	0.754
head + highest single	3	0.757	0.484	0.500	0.421
Hill, titration fine-tune	3	0.954	0.764	0.683	0.626
Hill + Loewe + dose gain	3	0.904	0.686	0.632	0.767
head + dose gain	3	0.898	0.676	0.568	0.865
displacement + dose gain	3	0.559	0.125	0.203	0.937
Hill + Bliss, zero-shot	3	0.977	0.791	0.529	0.591
Hill + Loewe, zero-shot	3	0.921	0.677	0.272	0.578
head + Bliss, zero-shot, 3 screens	3	0.943	0.749	0.623	0.030

Table 4: Round eleven on the public side. The Vanacloig column is the compound-cold centered Spearman (gene-cold rows: over held-out genes); nested ridge on fingerprint counts reads 0.359. The held-out column is each auxiliary screen's centered Spearman on its held-out compounds, in the order Hoepfner / Hillenmeyer / Wildenhain, or on a held-out dose of a seen compound in the dose rows. Growth AUROC is beside each for the cost on the wet-lab side.

question	arm	folds	Vanacloig	held-out public	growth AUROC
operator	control: tokens attend to the genes	3	0.265	–	0.850
	attend, head reads the cell only	3	0.307	–	0.866
	attend, two medium layers	3	0.325	–	0.850
	displace the gene embeddings	3	0.300	–	0.849
	displace, head reads the cell only	3	0.276	–	0.863
public bar	two CGT layers	3	0.333	–	0.885
	dropout 0.4	3	0.285	–	0.877
	weight decay 1e-3	3	0.275	–	0.859
	learning rate 5e-4	3	0.273	–	0.859
	graph prior 0.1	2	0.122	–	0.877
	graph prior 1	2	0.224	–	0.875
sources	Vanacloig, Hill + Loewe	3	0.350	–	0.913
	+ Hoepfner	3	0.299	0.039	0.917
	+ Hillenmeyer	3	0.333	0.008	0.909
	+ Wildenhain	3	0.254	0.333	0.920
	+ Hoepfner + Hillenmeyer	2	0.316	0.098 / 0.040	0.923
	+ all three	3	0.200	0.106 / 0.036 / 0.136	0.917
held-out dose	no dose channel	3	0.329	0.258	0.876
	FiLM	3	0.303	0.252	0.659
	dose gain on the offset	2	0.251	0.210	0.844
	Hill, no dose channel	2	0.303	0.220	0.881
	Hill + FiLM	1	0.177	0.282	0.848
	Hill + dose gain	3	0.241	0.214	0.876
gene-cold	attend	3	-0.001	–	0.826
	attend + pair term	3	0.007	–	0.851
	displace	3	0.002	–	0.826
	displace + pair term	3	0.003	–	0.825

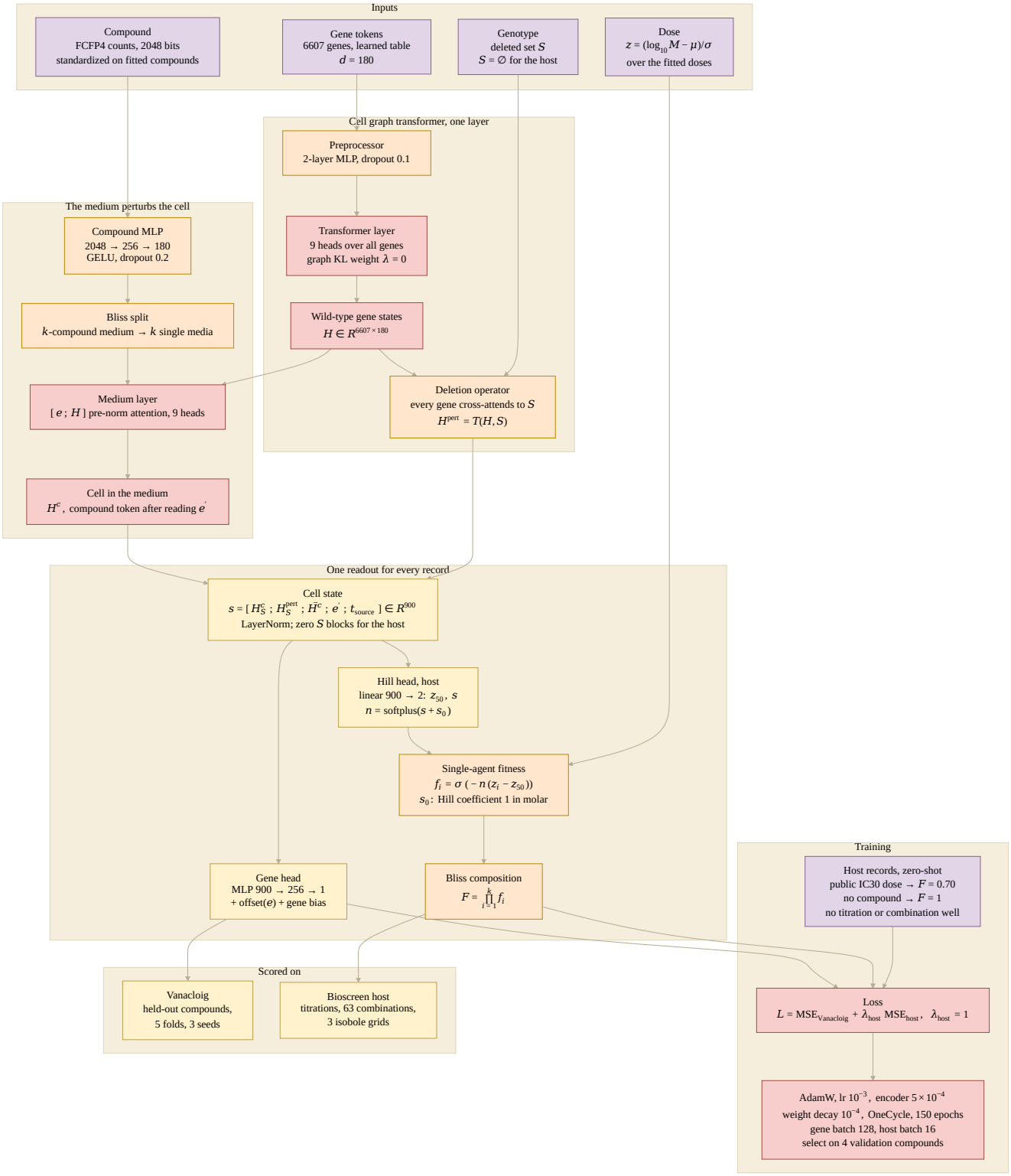


Figure 14: Supplement nine: the leading round-eleven arm, wired. The zero-shot Hill and Bliss arm of fig. 13. Purple, inputs and the host records; orange, the operators (the deletion operator, the compound MLP, the Bliss split, the Hill curve and its composition); red, the transformer layers and the optimization; yellow, the readouts. Every record is a cell: a published strain is read at its deleted genes, and the host is the record with no deleted gene, whose two gene blocks of the cell state are zero. The host's fitness is a Hill curve per inhibitor, its IC50 and slope read from the same cell state the gene head reads, evaluated at the medium's dose and multiplied over the inhibitors of a mixture; the only host records are the published IC30 anchors and the compound-free medium. The graph-prior weight is zero, so the nine gene graphs carry no signal into this arm.